Synthese

Is Time Dilation ‘Real’ ? Einstein and the Transverse Doppler Effect

--Manuscript Draft--

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Abstract:	As early as 1907, Einstein realized that the second-order transverse Doppler redshift reported by Stark in the spectral lines of fast-moving ions in canal rays could serve as a direct experimental test of time dilation. He rejected Stark’s dynamical explanation of the effect as a real frequency change of the moving ions. Instead, assuming that ions of the same kind emit the same spectral lines when at relative rest, Einstein offered a kinematical explanation of the frequency shift as a consequence of time dilation. He labeled the effect apparent, since it disappears for the co-moving observer. At the same time, he regarded it as real, in the sense that it would not occur if the old kinematics held. From the 1920s onward, Einstein argued that the behavior of atomic clocks calls for a dynamical explanation. The present paper shows, however, that even if an special relativistic explanation of the spectral identity of atoms were available, it would merely reinforce the claim that relativistic time dilation admits a purely kinematical interpretation. It concludes that modern dynamists misread Einstein’s plea for a dynamical account of clock behavior as a demand for explanation of the time dilation, whereas it was primarily motivated by the problem of its confirmation.

Response to the Referees for SYNT-D-26-00214

Response to reviewers' comments

I would like to thank the reviewers for their time and effort in reviewing the manuscript. The paper has been revised carefully in light of their comments.

The principal revisions concern mainly three issues: (a) the distinction between the 'real' and 'apparent' character of relativistic effects, following Einstein's terminology; (b) the clarification of the sense in which the transverse Doppler effect 'vanishes' for a co-moving observer; (c) the historical meaning of dynamical explanations within Einstein's work and their broader meaning as an autonomous theoretical program.

All changes have been tracked and are color-coded for your convenience. A version of the paper without track changes has been included for comparison.

Reviewer 1

Reviewer 2

Reviewer (2A) — First, it is claimed that length contraction and time dilation, in the eyes of Einstein, are "apparent" effects because they "vanish" for the co-moving observer, but are nonetheless "real" since they "cannot be eliminated in all frames at once" (pp. 2, 22, footnote 8). This latter criterion of reality is almost surely NOT what Einstein had in mind.

Reply: My explanation of the distinction between *real* and *apparent* was unclear. It is introduced as an exegetical reconstruction of Einstein's own usage. The clearest formulation occurs in his 1915 letter to Lorentz:

> Regarding the erroneous view that the Lorentz contraction was 'merely apparent,' [*scheinbar*] I am not free from guilt, without ever having myself lapsed into that error. It is real [*wirklich*], i.e., measurable with rods and clocks, and at the same time apparent [*scheinbar*] to the extent that it is not present for the co-moving observers (Einstein to Lorentz, Jan. 23, 1915; CPAE, Vol. 8, Doc. 47).

Accordingly, the paper does not claim that an effect is real because it *cannot be eliminated in all frames at once* (this applies to cases like the Ehrenfest paradox, see 2B). Rather, it reconstructs Einstein's own characterization:

– apparent = absent for the co-moving observer.

– real = experimentally measurable,

The transverse Doppler effect illustrates precisely this point. A spectrometer co-moving with the emitting ion records its proper frequency, so that no shift is observed in that experimental arrangement. A spectrometer in relative motion, by contrast, measures the shifted frequency. The effect is therefore relational, but this is precisely what Einstein meant by calling it simultaneously *real* and *apparent*.

The fact that the shift disappears in one observational arrangement does not make it illusory; it remains experimentally measurable in another (see also my response to 2C).

As I mentioned in a footnote, the distinction may also be illustrated by the familiar case of stellar aberration. The displaced position of the star is *apparent*, since it depends on the observer's state of motion. Yet aberration is *real* in an operational sense. If an astronomer points a telescope toward the star's true position rather than its aberrated position, the starlight will strike the wall of the telescope tube instead of reaching the eyepiece.

Reviewer (2B) — Applications of length contraction and time dilation can, in the appropriate circumstances, involve non-classical empirical predictions, such as the breaking of the string in Bell's two-rockets thought-experiment (footnote 1), stresses in a rotating disk (p. 2), the experimental confirmation of the transverse Doppler shift (section 2), and time retardation in the twin's "paradox" scenario (not mentioned in the paper). These predictions of special relativity are ones that ALL observers will agree on, including the observer in the relevant co-moving frame(s). It is in this sense that length contraction and time dilation are supposedly real. Indeed in footnote 1, the author explicitly states that the examples of "stresses and the possible structural failure of the rope (or of the rotating disk)" mean that length contraction is physically real for Einstein — these are coordinate-independent predictions.

Reply: I agree that the original formulation did not sufficiently distinguish the local and global senses in which a relativistic effect may be said to 'disappear' for a co-moving observer.

In the local case, a particular length contraction or time dilation can be made to disappear by passing to the rest frame of the relevant material element or clock. In that operational sense, the effect is 'apparent': the co-moving observer measures the proper length, proper time, or proper frequency. It is nevertheless 'real' in Einstein's sense because it is measurable using rods and clocks in a non-co-moving frame. The transverse Doppler effect should be understood in this context.

In extended or non-inertial situations, however, the local rest frames need not fit together into a single global co-moving inertial frame. This is what happens in the rotating disk: each element of the rim has an instantaneous rest frame, but these frames cannot be combined into a single global rest frame for the disk. In the original paper, I mentioned the Ehrenfest disk not as a general criterion for the reality of Lorentz contraction, but to recall the historical context in which the debate over whether Lorentz contraction is 'real' or merely 'apparent' first became explicit. In this correspondence, Einstein invokes the rotating disk to show that Lorentz contraction is a kinematical effect and therefore 'apparent' in the sense that it is absent for suitably co-moving observers. At the same time, it is 'real' in the sense that such contractions cannot be made to disappear globally for all inertial observers simultaneously, a fact manifested by the stresses that arise in the rotating disk (Einstein to Varićak, Mar. 3, 1911; CPAE, Vol. 5[10], Doc. 257a).

I have now removed the reference to the Ehrenfest paradox, as I agree that it was potentially misleading.

Reviewer (2C) — The transverse Doppler effect concerns the claim is that the frequency of emission of a moving ion, the emission being in the transverse direction, is different from the proper frequency as defined relative to the co-moving frame (which is the same as the proper frequency defined relative to K). It is arguably misleading to say that the effect "vanishes" relative to the co-moving frame. The effect might be described as a relation between two inertial frames, as is brought out on p. 16. Better put, it is a relation between proper frequency relative to K and that related to an ion moving relative to K. In the author's scenario, relative to K' this relation does not exist— there is a single "clock" and one cannot even define the Doppler shift, and so it cannot have zero value (if that is what the author means by "vanishing"). However, in the reversed scenario in which the observer at rest in K' is looking at emitting ions at rest relative to K, then the transverse Doppler shift is defined and the effect is the same as in the original case.

Reply:

I did not realize that my use of the term 'vanish' was indeed ambiguous. By saying that the effect 'vanishes' for the co-moving observer, I use the term in an 'operational' sense. The relevant question is what is recorded by measuring devices at rest with the system under investigation. A rod measured by co-moving rods has its proper length; a clock compared with co-moving clocks has its proper rate; an emitting ion observed by a co-moving spectrograph displays its proper frequency. In such an arrangement, the observable difference that defines the effect in a non-co-moving frame is not registered.

Thus, in the transverse Doppler case, the claim is not that the Doppler shift is an intrinsic property of the ion which somehow takes the value zero in its rest frame. Rather, a spectrograph co-moving with the emitting ion records the normal rest line, $\nu = \nu_0$, whereas a spectrograph in transverse relative motion records the shifted line, $\nu = \gamma^{-1}\nu_0$. This is the precise sense in which the shift 'vanishes' for the co-moving observer. It is absent in the co-moving observational arrangement (apparent), but experimentally measurable in the non-co-moving one (real).

Reviewer (2D) — The second point I want to make is more important. The author attempts to show, through an analysis of an important historical episode related to the testing of time dilation, that there is a confusion in the claim by Brown and others that such a "kinematical" effect (to use Einstein's original wording) has a dynamical underpinning, despite Einstein's belated recognition, endorsed by the author, that rods and clocks are physical systems governed by dynamical laws. Essentially, the author's argument is this. A dynamical account of, specifically, the transverse Doppler effect in special relativity, under plausible assumptions, would explain why all atomic clocks are identical when compared in the same rest frame, but no more. Explaining the transverse frequency shift requires, according to the author, more than that: it further requires time dilation regarded as a kinematical, not dynamical effect (pp. 3, 24).

Here are some objections to this view.

Bell showed in his famous 1976 paper on teaching special relativity that a purely dynamical (pre-quantum) model of a moving single-electron atom has time dilation unequivocally emerging from dynamical assumptions (largely Maxwell electrodynamics) in this specific model. Recall his point: the special merit of this approach "is to drive home the lesson that the laws of physics in any one reference frame account for all physical phenomena, including the observations of moving observers" (Bell 1976). The author's appeal to irreducible kinematics in fully accounting for the transverse Doppler effect seems to be in direct conflict with this important insight.

Furthermore, Bell makes a nice analogy with the parallel roles of thermodynamics (which gave Einstein the methodological template for his 1905 relativity paper) and statistical mechanics. He wrote:

"If you are, for example, quite convinced of the second law of thermodynamics, of the increase of entropy, there are many things that you can get directly from the second law which are very difficult to get directly from a detailed study of the kinetic theory of gases, but you have no excuse for not looking at the kinetic theory of gases to see how the increase of entropy actually comes about. In the same way, although Einstein's theory of special relativity would lead you to expect the FitzGerald contraction, you are not excused from seeing how the detailed dynamics of the system also leads to the FitzGerald contraction." (Bell 1976, as referenced in the paper.)

Of course the same can be said of time dilation. The key point is this. Naturally, Einstein used time dilation as established in the "kinematical" part of his 1905 paper in his treatment of the transverse Doppler effect. This was because the full dynamical theory of moving Ions and their emission properties was not available to him, and even today would be very complicated.

The beauty of the kinematical approach was that it did not rely on the fine-grained details of the dynamics, as the author acknowledges. It is analogous to the way thermodynamics leads to an entropy non-decrease law without specifying all the complicated details that would be required in a purely mechanical analysis of the relevant system. But as the current author concedes, special relativity is a set of constraints on theories. Indeed, it is much the way that thermodynamics is. Because something is hard to prove in what Einstein called a constructive (dynamical) theory of matter and radiation — even if it available —, while at the same amenable to a simple principle-theory treatment, does not mean that the latter is fundamental, or in principle irreplaceable. The author takes the lesson of the transverse Doppler effect to be that "kinematic and dynamics, from a strictly logical point of view, cannot be confronted with experience separately but only as a whole. From the point of view of explanation, however, the division of labour within this whole remains intact." (pp. 23, 24). The analogous, erroneous, claim would be that certain phenomena in the behaviour of gases, for example, can only be explained in principle by an irreducible combination of thermodynamics and statistical mechanics.

Reply: O thank the referee for raising this important point. The point of the paper is not to deny that Bell's program provides an independent theoretical motivation for a dynamical approach to relativity. Rather, it is to question a specific historical inference: namely, that Einstein's appeal to a dynamical theory of rods and clocks shows that he regarded time dilation itself as standing in need of a dynamical explanation.

On the interpretation defended in the paper, Einstein's point was different. A dynamical theory of matter is needed to account for the physical reliability of clocks, and in the case discussed here for the spectral identity of identical atoms. Once this identity has been secured, however, the transverse Doppler shift is explained kinematically, as a consequence of the Lorentz transformation of time. In this sense, the effect is 'apparent' in Einstein's terminology, though empirically 'real.' A dynamical explanation of the shift itself would seem to require the Lorentz–Ives strategy of treating the moving atom as undergoing a real change of intrinsic frequency.

Einstein's plea for a dynamical theory of rods and clocks has often been interpreted as a demand for *explanation* of the new kinematics, whereas it was primarily motivated by the problem of its *confirmation*. The Lorentz transformations define the kinematical structure of special relativity. In order to test them empirically, one must employ rods and clocks as linked to the coordinate system. Yet rods and clocks are themselves physical systems governed by dynamical laws, which are in turn required to invariant with respect to the Lorentz transformations. For this reason, kinematics and dynamics, from a strictly logical point of view, cannot be confronted with experience separately, but only as a whole.

In the new version of the paper, I point out explicitly that the dynamical approach can be pursued with relying on Einstein's authority by following Bell's strategy. This is however an independent philosophical development, not evidence for the historical claim that Einstein himself sought a dynamical explanation of time dilation. I ultimately decided to relegate a brief comment on Bell to a footnote, as introducing such a discussion in the main text seemed too abrupt.

Reviewer (2E) — It stated that the Lorentz transformations make their "perfect symmetry in x and t explicit". Is it perfect?

Reply: I agree that the formulation was too strong. I have replaced it with a more cautious statement: the Lorentz transformations make explicit the reciprocal mixing of spatial and temporal coordinates, especially when the time coordinate is written as ct.

Reviewer (2F) — It is claimed that Einstein was the first to see time dilation as a genuinely new physical effect. But Larmor and Lorentz independently had a clear premonition of time dilation in 1897 and 1899 respectively, though it is not clear that they thought of it as a universal property of ideal clocks.

Reply: I thank the referee for this correction. I have qualified the claim. The revised text now acknowledges that Larmor and Lorentz had obtained related formal results before Einstein. The point I intended to make is narrower: Einstein's distinctive contribution was to interpret clock retardation as a universal consequence of the new kinematics, applying to any suitable clock, rather than as a feature of a particular electromagnetic model.

Reviewer (2G) — The coordinate time interval in K between two successive ticks of a moving clock cannot be measured by a single clock at rest in K, since the ticks occur at different spatial points in K. This risks inconsistency with the claim that the moving clock runs "more slowly" than an identical clock at rest in K.

Reply: I agree. I have revised the relevant passage to make explicit that Δt is read off by the synchronized lattice of clocks at rest in K, whereas $\Delta t'$ is the proper time measured by the single clock at rest in K'. I have also qualified the potentially misleading phrase that the moving clock "runs more slowly" (used by Einstein) with a formulation in terms of the longer coordinate-time interval assigned in K to successive ticks of the moving clock.

Reviewer (2H) — The manuscript seems to treat the invariance of the number N of oscillations as a substantive assumption based on the relativity principle. But N is a count and is independent of all coordinate frames by its nature.

Reply: I agree that the previous wording was misleading. I have revised the manuscript throughout to distinguish the trivial invariance of the count N from the substantive physical premise. The point is not that N itself is made invariant by the relativity principle, but that uniform motion is assumed not to alter the intrinsic oscillatory process of the emitter. Thus identical ions, when compared at rest, possess the same proper frequency $\nu_0 = N/\Delta t'$ in the co-moving frame. The transverse Doppler effect then arises because the same count N is referred to different time intervals, $\Delta t'$ and $\Delta t = \gamma \Delta t'$, in the non-co-moving frame.

Reviewer (2I) — Ritz is described as criticizing the abandoning of absolute time and rigid bodies in the Lorentz–Einstein theory. But this theory does not abandon idealized rigid bodies.

Reply: I agree. I have revised the wording to clarify that Ritz was criticizing the abandonment of absolute time and the classical notion of absolutely rigid bodies, not the use of idealized rods or rigid bodies as measuring instruments within the relativistic framework. More specifically, Ritz had in mind something similar to the contrast between Abraham's 'rigid' electron and the 'deformable' electron of Lorentz's and Einstein's theories.

Reviewer (2J) — Typos: first line p. 16; third line final paragraph p. 18; equation in last line of first paragraph of section 4, p. 20.

Reply: I thank the referee for pointing these out. I have corrected these typographical errors in the revised manuscript.

Reviewer 3

Reviewer (3A) — The present submission contributes to the literature on the relation between dynamical and kinematical considerations in special relativity. Specifically, it disputes a narrative to the effect that Einstein initially treated time dilation as a merely apparent coordinate artifact but sought a dynamical explanation on which time dilation reflects real changes, thereby suggesting a precursor to today's dynamical view. To challenge this narrative, the paper reconstructs in detail the role of the kinematical/dynamical distinction in Einstein's treatment of the transverse Doppler effect. It argues that Einstein was interested not in a dynamical explanation of the transverse Doppler effect but rather in a dynamical backing for assumptions going into experimental testing of the effect.

This paper is timely because it concerns debates over kinematical and dynamical explanation in special relativity. The topic is also relevant to those with historical interests in the special theory of relativity. The historical analysis exhibits detailed engagement with the primary sources and substantial work in synthesizing them. It successfully makes its conceptual case on the basis of this analysis.

Minor typos: 2: "as [shown] by the stresses in Ehrenfest's disk" 9: "[in] November of 1905" 23: "which are in turn required to invariant with [respect to] the Lorentz transformations"

Reply: I have corrected the remaining typos wherever the corresponding phrases still appear in the revised version.

Is Time Dilation 'Real'? Einstein and the Transverse Doppler Effect

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Abstract

As early as 1907, Einstein realized that the second-order transverse Doppler redshift reported by Stark in the spectral lines of fast-moving ions in canal rays could serve as a direct experimental test of time dilation. He rejected Stark's *dynamical explanation* of the effect as a real frequency change of the moving ions. Instead, assuming that ions of the same kind emit the same spectral lines when at relative rest, Einstein offered a *kinematical explanation* of the frequency shift as a consequence of time dilation. He labeled the effect *apparent*, since the shifted line is not recorded by a spectrograph co-moving with the ion. At the same time, he regarded it as *real*, in the sense that it would not occur if the old kinematics held, and is thus empirically testable. From the 1920s onward, Einstein argued that the behavior of atomic clocks calls for a dynamical explanation. The present paper shows, however, that even if a special relativistic explanation of the spectral identity of atoms were available, it would merely reinforce the claim that relativistic time dilation admits a purely kinematical interpretation. It concludes that modern dynamicists misread Einstein's plea for a dynamical account of clock behavior as a demand for an *explanation* of time dilation, whereas it was primarily motivated by the problem of its *confirmation*.

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Is Time Dilation 'Real'? Einstein and the Transverse Doppler Effect

Abstract

As early as 1907, Einstein realized that the second-order transverse Doppler redshift reported by Stark in the spectral lines of fast-moving ions in canal rays could serve as a direct experimental test of time dilation. He rejected Stark's *dynamical explanation* of the effect as a real frequency change of the moving ions. Instead, assuming that ions of the same kind emit the same spectral lines when at relative rest, Einstein offered a *kinematical explanation* of the frequency shift as a consequence of time dilation. He labeled the effect *apparent*, since the shifted line is not recorded by a spectrograph co-moving with the ion. At the same time, he regarded it as *real*, in the sense that it would not occur if the old kinematics held, and is thus empirically testable. From the 1920s onward, Einstein argued that the behavior of atomic clocks calls for a dynamical explanation. The present paper shows, however, that even if a special relativistic explanation of the spectral identity of atoms were available, it would merely reinforce the claim that relativistic time dilation admits a purely kinematical interpretation. It concludes that modern dynamicists misread Einstein's plea for a dynamical account of clock behavior as a demand for an *explanation* of time dilation, whereas it was primarily motivated by the problem of its *confirmation*.

Keywords: Einstein, Special Relativity, Time Dilation, Transverse Doppler Effect, Confirmation, Explanation

Introduction

Over the last two decades, philosophical discussions of special relativity have repeatedly returned to the question of whether relativistic effects, most notably length contraction and time dilation, ultimately call for a *dynamical* explanation (Brown, 2005; Brown and Pooley, 2006), a *kinematical* explanation (Norton, 2008; Janssen, 2009), or whether one should move beyond this binary choice altogether (Acuña, 2016). While the debate is primarily theoretical in intent (Read, 2020; Acuña and Read, 2026), proponents of the dynamical approach have often framed it against a historical background. In particular, it has been argued that, although Einstein initially presented length contraction and time dilation as merely *apparent* coordinate effects, he was ultimately aiming at an account in which these effects reflect *real* physical changes in the equilibrium configurations of moving atomic systems (Brown and Read, 2022). Indeed, he later acknowledged that rods and clocks should have been treated as solutions of the underlying dynamical laws (Einstein, 1949a, 61). It has therefore been suggested that Einstein himself may, at least retrospectively, be seen as gesturing toward a dynamical approach (Brown and Read, 2022, 70).

This paper aims to further challenge this historical narrative, in the hope that this result will bear directly on contemporary debates concerning the theoretical role of the dynamical–kinematical distinction. As recent literature has shown (Giovanelli, 2023), concerns about the proper use of the categories 'real' and 'apparent', as well as 'kinematical' and 'dynamical', had already emerged in the early 1910s. Reacting to Ehrenfest's (1909) paradox, some contemporary relativists (Ignatowski, 1910; Varićak, 1911) argued that stresses in a rotating disk would arise only if 'Lorentz contraction' were interpreted as a *real* dynamical effect, produced by molecular forces within the material of the rod. By contrast, 'Einstein contraction' was said to be merely an *apparent* coordinate effect resulting from an arbitrary synchronization of clocks. It could therefore be made to disappear by adopting a different synchronization convention. Relativistic length contraction, Varićak argued, "is only a psychological and not a physical fact, *i.e.*, the body has not really undergone any change" (Varićak, 1911, 169).

Einstein (1911) felt compelled to take a public stance against this view. Length contraction is indeed *apparent* insofar as it is absent for observers co-moving with the rod; yet it is nevertheless *real* in Einstein's own sense, namely, measurable by means of rods and clocks. Consider a rod AB at rest in K' and moving relative to K. Its *projection* in K is the segment $A'B'$ determined by the positions A' and B' of the endpoints A and B, measured simultaneously in K. Relativity theory predicts that $A'B' < AB$. In principle, this prediction could be confirmed or disconfirmed by measuring AB and $A'B'$ with the same unit rod, first while it is momentarily co-moving with AB and then after it has been brought to rest in K without any alteration of its length. As Einstein put it to Varićak: "The contraction can be ascertained by measurement, *i.e.*, it is 'real'" (Einstein to Varićak, Mar. 3, 1911; Einstein, 1987, Vol. 5[10], Doc. 257a).

This paper argues that the same dialectic between the *apparent* and the *real* emerges more clearly in the case of time dilation. Like length contraction, time dilation is 'apparent' insofar as a clock at rest in its own frame records its proper rate; yet it is 'real' insofar as it has empirically testable consequences.[1] In contrast to length contraction, however, the empirical testability of time dilation soon appeared to be practically feasible. As early as 1907, Einstein appealed to Johannes Stark (1906)'s experimental work on fast-moving ions in canal rays, which seemed to suggest the presence of a second-order Doppler effect in the transverse direction of motion, that is, even when the classical longitudinal Doppler effect disappears. A concrete possibility for testing relativistic kinematics was thus at hand (Giuliani, 2013).

As the paper will show, from the outset, Einstein (1907b) rejected Stark's (1906) original explanation, according to which ions in motion were thought to undergo a 'real', dynamical contraction of their intrinsic frequency. Spectroscopy showed that atoms and ions of the same species always display the same characteristic spectral lines when compared at rest, independently of their prior acceleration history. The

[1] In what follows, I use the term 'kinematics' and avoid explicit reference to the 'geometry' of spacetime, since Minkowski's framework was far from being widely adopted in the period discussed in this paper.

principle of relativity implies that atoms of the same kind realize the same intrinsic frequencies in all inertial systems. In the transverse Doppler configuration, there is no relative motion along the line of sight, and hence no classical Doppler contribution. Einstein could then conclude that the reduced frequency must therefore be ascribed to the relativistic relation between time coordinates in different inertial frames, that is, to time dilation itself. Accordingly, Einstein initially labeled the change in the observed frequency 'apparent' in order to emphasize that the intrinsic frequency of ions is supposed to be invariant (Einstein, 1908, 422; Einstein, 1910, 133). Nevertheless, the transverse Doppler effect is 'real' in Einstein's sense, since, if the new kinematics holds, a non-co-moving spectrograph would measure the shifted spectral lines, whereas a co-moving spectrograph would measure the ion's proper frequency. The transverse Doppler effect thus offered, at least in principle, a direct test of time dilation, provided that atomic spectral emitters can serve as 'good' clocks.

From the 1920s onward, Einstein explicitly emphasized that the behavior of atomic clocks could not yet be derived from a complete microscopic theory (Einstein, 1921, 1923, 1924, 1926, 1949a). He conceded that, in a fully developed theory, the existence and behavior of clocks would, in principle, emerge as consequences of the fundamental equations themselves. Any theory capable of supplying oscillatory processes usable as clocks would have to single out solutions characterized by a definite, invariant period, determined by the structural constants of the theory. If the theory were Lorentz invariant, such solutions would recur in all inertial frames in the same way, thereby guaranteeing the identical behavior of clocks when compared at rest. From a modern perspective, an example of such an invariant scale is provided by a mass parameter in relativistic quantum theory, which fixes a characteristic frequency, the Compton frequency $\nu_C = mc^2/h$.

Such a theory would then indeed provide a *dynamical explanation* of why all atomic clocks are *identical* when compared in the same rest frame and therefore emit the same spectral lines. However, this line of reasoning leads to an inevitable conclusion. Given the spectral identity of atoms, and assuming that there is no contribution from motion along the line of sight, the transverse frequency shift must inevitably admit a purely *kinematical explanation*, namely as a consequence of time dilation itself. Somewhat paradoxically, the full realization of the dynamical program shows that time-dilation effects, including the transverse frequency shift, are purely kinematical. In this sense, this paper suggests that the transverse Doppler effect offers a particularly illuminating case for reassessing the long-standing debate between kinematical and dynamical interpretations of special relativity.

In particular, the paper concludes that much of the misunderstanding arises from the fact that Einstein's demand for a dynamical account of rods and clocks has been read as a demand for an *explanation* of the new kinematics, whereas it was primarily motivated by the problem of its *confirmation*. Einstein never tired of emphasizing that relativistic kinematics was not merely conventional but, in principle, testable by means of rods and clocks. Yet rods and clocks are complex physical systems governed by dynamical laws that are themselves supposed to conform to relativistic kinematics. From a logical point of view, kinematics and dynamics can receive empirical confirmation only as a whole. Nevertheless, within this whole, dynamical and kinematical explanations remain distinct. The paper argues that the transverse Doppler effect brings this out in especially vivid terms: dynamics explains the sameness of clocks' frequencies when compared at rest, while kinematics alone explains the frequency shift across inertial frames.

1 Time Dilation and the Transverse Doppler Effect in Einstein's 1905 Relativity Paper

Although Einstein's 1905b relativity paper is among the most cited in the history of science, a brief review of its basic structure may nevertheless be useful for locating the respective derivations of time dilation and the transverse Doppler effect. Starting from the two postulates—the relativity principle and the light postulate—the first, kinematical part of the paper aims to develop a new kinematics of "rigid bodies (coordinate systems), clocks, and electromagnetic processes" (Einstein, 1905b, 892) that resolves their apparent incompatibility. In §1, by introducing the synchronization of clocks using light rays, Einstein shows that there is no *a priori* reason to assume that $t' = t$, that is, that two events which are simultaneous with respect to K' must also be simultaneous with respect to K. In §2 he shows that, as a consequence, there is no *a priori* reason to maintain the classical identification $x' = x$. Since spatial distances depend on the adopted notion of simultaneity, the length of a moving rod as measured in the co-moving system K' is not necessarily equal to the rest length of that rod as measured in the rest frame K.

Once these two prejudices are abandoned, it becomes possible to consider linear transformations between coordinate systems in uniform parallel translation that differ from the classical ones, for which $t' = t$ and $x' = x - vt$. In §3 Einstein then seeks a new set of coordinate transformations that leave the velocity of light c invariant. The derivation is somewhat cumbersome (Martinez, 2009, 320–354), but Einstein nevertheless arrives at the so-called Lorentz transformations for motion along the common x–x' axis:

$$x' = \gamma(x - vt), \qquad t' = \gamma\Big(t - \frac{v}{c^2}x\Big), \qquad \gamma = \frac{1}{\sqrt{1 - \frac{v^2}{c^2}}}.$$

The transformation appears here for the first time in its modern form, which makes explicit the reciprocal mixing of the spatial and temporal coordinates, especially when time is written as ct.[2]

Finally, in the concluding sections of the kinematical part, Einstein derives the physical consequences of the Lorentz transformations. Once coordinates are interpreted as being measured by means of rods and clocks, the theory yields specific predictions about the behavior of such systems that are, in principle, accessible to experiment. In §4 he derives both length contraction and time dilation. In §5 Einstein then derives the relativistic law of velocity addition, which resolves the tension between the principle of relativity and the light postulate. While length contraction had already been discussed

[2] Although Einstein originally used β to denote the Lorentz factor, in what follows I adopt the modern convention, reserving γ for the Lorentz factor (γ^{-1} for the inverse Lorentz factor) and β for the dimensionless velocity $\beta = v/c$.

in Lorentz's earlier work, time dilation appears here as a genuinely new physical effect. Although Joseph Larmor and Lorentz had already obtained related formal results (Brown, 2005, sec. 4.4 and 4.5), Einstein gave clock retardation its distinctive physical interpretation as a universal consequence of the new kinematics, applying to any suitable clock. Indeed, notwithstanding the well-known priority disputes surrounding the 'discovery' of the special theory of relativity, the argument can be made that the introduction of time dilation as a universal effect constitutes Einstein's distinctive contribution (Rindler, 1970). It would therefore be appropriate—although this is not customary—to speak of 'Einstein dilation' as the natural counterpart to 'Lorentz contraction'.

1.1 Time Dilation and the Relativistic Clock Retardation

As one might expect, the young Einstein's derivation of time dilation relies purely on algebraic manipulation.[3] He considers a clock that is permanently at rest in the moving system K'. This means that its spatial coordinate in K' is constant and, in particular, that $x' = 0$ for all times. The clock, however, moves with respect to K with constant velocity v along the x-axis. Einstein therefore makes implicit use of the Lorentz transformation relating the spatial coordinates of K and K'.

$$x' = \gamma(x - vt),$$

Since $x' = 0$, this condition immediately yields

$$0 = \gamma(x - vt)\,.$$

Formally, a product vanishes if and only if at least one of its factors vanishes. Since $\gamma \neq 0$, it follows that

$$x = vt\,,$$

which is simply the equation of uniform motion with velocity v in the system K: a clock at rest in K' moves uniformly with velocity v in K. The time indicated by this clock is denoted by t', which is related to t by the Lorentz time-transformation equation:

$$t' = \gamma\Big(t - \frac{v}{c^2}x\Big)\ .$$

Substituting $x = vt$, one obtains

$$\begin{aligned} t' &= \frac{1}{\sqrt{1-\frac{v^2}{c^2}}}\left(t - \frac{v^2}{c^2}\,t\right) \\ &= \frac{1}{\sqrt{1-\frac{v^2}{c^2}}}\left(1 - \frac{v^2}{c^2}\right)t \\ &= t\sqrt{1-\frac{v^2}{c^2}}. \end{aligned}$$

[3] This is one of the reasons I prefer to avoid the adjective 'geometrical' in this context.

It follows that the coordinate time interval Δt, read off by the synchronized lattice of clocks at rest in K between two successive ticks of a clock moving with respect to K, is longer by a factor γ than the proper time interval $\Delta t'$ measured by that same clock in its co-moving system K'.

Einstein suggests, albeit rather implicitly, that for velocities small compared with c, one may expand the square root in a Taylor series about $v/c = 0$ (i.e., the Maclaurin series in the dimensionless parameter $(v/c)^2$):

$$\sqrt{1-\frac{v^2}{c^2}} = 1 - \frac{1}{2}\frac{v^2}{c^2} - \frac{1}{8}\frac{v^4}{c^4} - \cdots .$$

Keeping only the leading correction and neglecting terms of fourth and higher order in v/c, one obtains

$$1 - \sqrt{1-\frac{v^2}{c^2}} \approx \frac{1}{2}\frac{v^2}{c^2}.$$

Hence the retardation per unit time is approximately

$$\frac{\Delta t}{t} \approx \frac{1}{2}\left(\frac{v}{c}\right)^2.$$

The longer the clocks run, the larger the difference between their readings as judged in the frame K. The retardation is second order in v/c. For ordinary velocities ($v \ll c$), the quantity v^2/c^2 is extremely small, so the retardation accumulates only very slowly. However, by expressing the effect *per second*, Einstein emphasizes that the deviation is systematic and cumulative. A precise quantitative prediction, however small, can be tested if either the velocity is sufficiently high, or the observation time is sufficiently long, or both.

1.2 The Longitudinal and Transverse Doppler Effects

After deriving "the required propositions of the kinematics that correspond to our two principles", Einstein proceeds "to show their *application* in electrodynamics" (Einstein, 1905b, 907). The sharp separation between the kinematical part of the paper and the subsequent dynamical parts (involving electromagnetism and particle dynamics) is one of the most characteristic features of Einstein's approach to the electrodynamics of moving bodies. The kinematical framework is established independently of dynamics, and in particular independently of Maxwell's equations; only afterward is it brought to bear on dynamical theories. The second part of the paper accordingly 'applies' the new kinematics to well-established dynamical laws in order to test whether they remain unchanged under the Lorentz transformations. The procedure consists in formulating the dynamical equations with reference to the 'moving' system K', performing the Lorentz transformation, and verifying whether the transformed equations retain the same form in the 'rest' system K. If they do not, the equations must be modified, and such modifications may entail empirically testable consequences.

In §6 Einstein shows that Maxwell–Hertz's equations already satisfy this criterion. From his point of view, the fact that no additional terms need to be introduced is a fortunate circumstance, since the Lorentz transformations were not derived from Maxwell's equations themselves. In §7 Einstein applies the same procedure to the

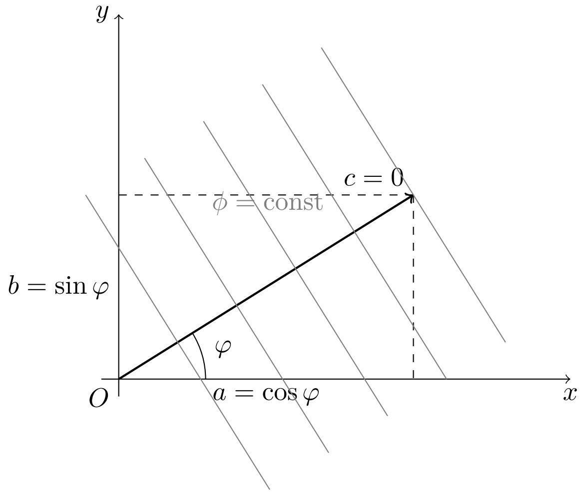

Figure 1: Plane wave propagating in the x–y plane

description of optical plane waves. The Lorentz invariance of the wave equation entails modified relations between frequency, direction of propagation, and relative motion. This yields a unified derivation of the aberration of light and the relativistic Doppler effect as consequences of the Lorentz transformations. Both aberration and the longitudinal Doppler effect were already known in classical optics; however, Einstein's treatment provides their exact relativistic form, valid to all orders in v/c, without recourse to ether-based dynamical hypotheses. Moreover, Einstein predicts the transverse Doppler effect, a genuinely new phenomenon absent from classical theory, which arises solely from time dilation and has no classical analogue.

Einstein's derivation proceeds as follows (Einstein, 1905b, 910–11). Consider a moving, non-accelerated system K' in which a very distant light source is at rest. In this system, the emitted radiation can locally be represented as a plane wave, that is, by sinusoidal electric and magnetic field components whose phase

$$\phi' = \omega'\left[t' - \frac{a'x' + b'y' + c'z'}{c}\right]$$

defines a family of parallel phase planes $\phi' = \text{const}$ propagating rigidly through space. Here $\omega' = 2\pi\nu'$ is the angular frequency in the rest frame of the source, and (a', b', c') are the direction cosines of the wave normal. Without loss of generality, if the wave propagates in the x–y plane, one may set $a' = \cos\varphi$, $b' = \sin\varphi$, and $c' = 0$, where φ is the angle between the direction of propagation and the x-axis.

According to the principle of relativity, the same physical wave must also be describable in the rest system K with respect to which K' moves with uniform velocity. Einstein assumes that the wave preserves its plane-wave character in K, so that it can again be represented by a phase of the same functional form,

$$\phi = \omega\left[t - \frac{ax + by + cz}{c}\right].$$

Since the wavefronts themselves are physical entities, one and the same wavefront must correspond to the same value of the phase in all inertial frames (Einstein, 1905b, 911). This requirement is expressed by the condition

$$\phi' = \phi.$$

To implement this condition, the phase ϕ' must be rewritten in terms of the unprimed variables x, y, z, t. That is, the mathematical description of the wave given in K' must be expressed using the coordinates of K. This is achieved by substituting the Lorentz transformations for the time coordinate and for the longitudinal coordinate x, together with $y = y'$ and $z = z'$, into the expression for ϕ'. Upon regrouping terms, the phase again retains the same functional form, but with transformed coefficients:

$$a = \frac{a' + \frac{v}{c}}{1 + \frac{v}{c}a'}, \qquad b = \frac{b'}{\gamma\big(1 + \frac{v}{c}a'\big)}, \qquad c = \frac{c'}{\gamma\big(1 + \frac{v}{c}a'\big)}, \qquad \omega = \gamma\, \omega' \Big(1 + \frac{v}{c}a'\Big)\,.$$

As mentioned, a, b, c are the direction cosines of the wave normal, that is, the components of the direction of propagation (fig. 1). Their transformation therefore expresses the change in the apparent direction of propagation between K' and K. By definition, the angular frequency ω of a plane wave is given by the coefficient of the time coordinate in the phase when the spatial coordinates are held fixed. Accordingly, the coefficient multiplying t in the transformed phase determines the observed angular frequency ω in the system K. The invariance of the phase thus entails two distinct but closely related effects. The transformation of the temporal coefficient yields a change in frequency, that is, the relativistic Doppler effect, whereas the transformation of the spatial coefficients encodes a change in the apparent direction of propagation of the wave, that is, the aberration of light. The Doppler effect and aberration therefore arise simultaneously from the same Lorentz transformation of the phase, as two aspects of a single kinematical procedure.

We are here interested in the frequency shift (Einstein, 1905b, 911). Transforming the phase of the wave from K' to K and passing from angular to ordinary frequency, $\nu = \omega/(2\pi)$, one obtains the relativistic Doppler formula

$$\nu = \nu' \, \frac{1 + \frac{v}{c}\cos\varphi}{\sqrt{1 - \frac{v^2}{c^2}}}.$$

For longitudinal propagation along the x–axis, one has $\varphi = 0$ and hence $\cos\varphi = 1$. The general expression then reduces to the familiar relativistic formula for the longitudinal Doppler effect,

$$\nu = \nu' \, \frac{1 + \frac{v}{c}}{\sqrt{1 - \frac{v^2}{c^2}}} = \nu' \sqrt{\frac{1 + \frac{v}{c}}{1 - \frac{v}{c}}}.$$

As this shows, when one starts from a source at rest in the moving system K' and evaluates the frequency in the system K, the longitudinal Doppler effect combines a classical Doppler factor, $1 + \frac{v}{c}$, associated with relative motion along the line of sight, with an additional relativistic correction $1/\sqrt{1-\frac{v^2}{c^2}}$. In classical wave theory, if the source is approaching the observer, the wave crests arrive more frequently, and the observed frequency is therefore higher than that of an identical source at rest. Relativity predicts a further frequency shift beyond this classical effect.

The case $\cos\varphi = 0$ corresponds to motion perpendicular to the direction of propagation of the light, that is, to transverse motion (along the y–axis in fig. 1). In this case the formula reduces to

$$\nu = \frac{\nu'}{\sqrt{1-\frac{v^2}{c^2}}},$$

showing that even when there is no component of the relative motion along the line of sight, the observed frequency nevertheless differs from the emitted one. A frequency shift therefore appears even though there is no classical Doppler effect in the Galilean theory of waves. This purely relativistic transverse Doppler effect has no analogue in pre-relativistic wave theory.

2 Testing the New Kinematics: Stark's Second-Order Doppler Effect

In the 1905 paper, time dilation and the transverse Doppler effect appear, respectively, among the direct and indirect consequences of the Lorentz transformations. Einstein does not explicitly relate the two effects to one another, nor does he discuss the possibility of an empirical test of the transverse Doppler effect. He likely did not think that wave sources were available that could attain velocities large enough for such a small effect to be experimentally accessible. Rather, in §8 of his relativity paper, Einstein derives the transformation law for the frequency of light and argues that the energy of light transforms in the same way as its frequency, i.e. $\frac{E'}{E} = \frac{\nu'}{\nu}$. On this basis, he derives the radiation pressure exerted by light on a perfectly reflecting mirror in uniform motion. This result is subsequently exploited in the September paper on the inertia of energy (Einstein, 1905a). In §9 Einstein then turns to the Maxwell equations with sources. The only explicit discussion of the empirical testability of the new theory occurs in §10 of the paper. In this section Einstein derives the equation of motion of the slowly moving 'electron', an expression that he uses to denote massive charged particles in general. Insofar as electrons (in the sense of elementary particles) function as fast-moving charged test particles, the predicted effects are, in principle, accessible to observation.

Indeed, Einstein's 1905 relativity paper was cited for the first time on November of 1905 by the Göttingen experimentalist Walther Kaufmann (1905, 954), in a short note presented at a plenary session of the Prussian Academy of Sciences. The note reported on Kaufmann's new experiments with Becquerel rays, or β-rays, emitted by radioactive substances. As is well known, Kaufmann's results appeared unfavorable to relativity, a conclusion he articulated more fully in a longer essay published at the beginning of 1906 (Kaufmann, 1906). The data aligned more closely with Abraham's theory, according to which the electron was assumed to be non-deformable. The issue thus seemed to be settled against the Lorentz–Einstein electron in favor of Abraham's

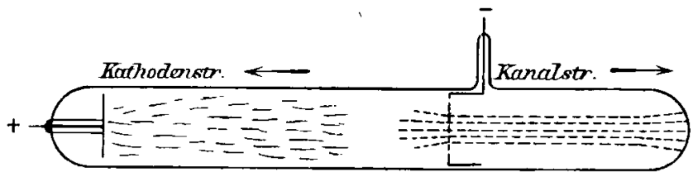

Figure 2: Discharge tube producing cathode rays and canal rays (Stark, 1906, 405)

spherical model. However, in March Planck (1906a) urged caution. The results of Kaufmann's experiments were not sufficiently precise to warrant a definitive refutation.

According to Planck (1906b), relativistic dynamics had the significant advantage of being derived without reliance on any specific model of the electron and was therefore worth exploring. At this juncture, the young Einstein himself clearly regarded experiments on fast-moving electrons as the only means of testing the theory. In August 1906 he submitted a paper (Einstein, 1906) proposing a new experimental method to determine the ratio of transverse to longitudinal electron mass at high velocities using electrostatic deflection alone. The paper was published in issue 21 of the *Annalen der Physik*. In the same issue, a contribution by Johannes Stark (1906), submitted in September 1906, also appeared. It is therefore plausible that Einstein read Stark's paper upon receiving his copy of the *Annalen* issue. He may then have realized that wave sources were indeed available to test the transverse part of the relativistic Doppler effect.

2.1 Stark's Canal-Ray Experiments and Velocity-Dependent Spectral Effects

Stark could rely on decades of spectroscopic work showing that excited gases and vapors do not produce continuous spectra, but rather sharp spectral lines appearing at well-defined positions in a spectrograph (Hentschel, 2002). The Zeeman splitting of such lines in a magnetic field was found to be consistent with a charge-to-mass ratio equal to that of the electron, and the sense of the splitting indicated a negative charge. Within the framework of Lorentz's electron theory, it was therefore concluded that the 'centers' (*Zentren*) of emission responsible for line spectra are negatively charged oscillating electrons (Stark, 1906, 401). Stark's central aim was to identify the physical systems that serve as the 'carriers' (*Träger*) of different spectra, that is, the material systems bearing the electronic oscillatory processes responsible for line emission (Stark, 1906, 402). He conceived the carrier as a positive atomic ion containing bound negative electrons executing elastic oscillations about equilibrium positions. Radiation was attributed to these accelerated charges, and spectral lines were identified with their characteristic oscillation frequencies; the sharpness and reproducibility of the lines were taken as empirical evidence for stable intrinsic atomic periodicities. The distinction between 'centers' and 'carriers' is crucial to Stark's theory, since it allows the physical state of the carrier to influence the behavior of the oscillating centers.

Stark's experimental investigations begin with a gas-filled discharge tube, typically containing hydrogen (fig. 2). When an electrical discharge is applied, the discharge current is largely carried by fast negative electrons, which are accelerated by the electric field toward the anode. As these electrons traverse the gas, they undergo collisions with neutral atoms or molecules. In sufficiently energetic collisions, electrons are knocked out of atoms, producing positive ions, that is, "atoms that have lost one or several negative electrons through ionization" (Stark, 1906, 402). These ions are accelerated toward the cathode through a potential drop V concentrated in the cathode fall, which sets the velocity scale of the moving ions. If the cathode is perforated, a fraction of the ions passes through the holes and continues beyond the cathode with approximately rectilinear motion, forming a beam propagating away from it. These beams of positive ions are known as canal rays (*Kanalstrahlen*), named after the 'canals' in the cathode through which they pass (Stark, 1906, 403).

When accelerated toward the cathode, the positive ions emit characteristic spectral radiation in all directions as a result of excitation processes in the discharge. Stark analyzed this radiation using a prism spectrograph. The emitted light entered the instrument through a narrow slit, which selected a well-defined direction of propagation (longitudinal or transverse with respect to the ion motion). After passing through the slit, the light is rendered approximately parallel and passed through a glass prism, which separated different wavelengths λ by refraction; the dispersed radiation is finally recorded on a photographic plate. In the case of hydrogen, Stark took the Balmer series as empirically given, that is, as a set of well-defined spectral lines Hα, Hγ, ... (Stark, 1906, 417). Whether the emitting system consisted of neutral hydrogen in a discharge tube or of hydrogen ions in canal rays, the same characteristic lines reappeared. When the emitting ions moved along the line of sight of the spectrograph (toward b or c in fig. 3), Stark observed that the entire pattern of spectral lines was displaced toward longer wavelengths (redshift) for receding motion (toward c), and toward shorter wavelengths (blueshift) for approaching motion (toward b). The magnitude of this Doppler displacement increased with the ratio v/c:

$$\frac{\Delta\lambda}{\lambda} \sim \pm\frac{v}{c}.$$

Stark concluded that sharp line spectra (*Linienspektrum*) are emitted by individual atomic ions in translational motion. By contrast, the *band spectrum* (*Bandenspektrum*)—consisting of groups of closely spaced lines forming characteristic bands—exhibited no measurable Doppler displacement. Stark therefore attributed band spectra to neutral atoms or molecular aggregates formed in recombination processes, whose translational velocities are much smaller than those of the canal-ray ions (Stark, 1906, 426).

In the first two parts of his paper, Stark established the ordinary first-order Doppler effect proportional to v/c and concluded that "[t]he band spectrum and the line spectrum of hydrogen do not have the same carrier" (Stark, 1906, 414). In part III, he raised a further question: whether an additional displacement of spectral lines might exist that depends only on the magnitude of the ion's velocity rather than on its direction. Any such effect would necessarily be proportional to v^2/c^2, since linear terms change sign when the direction of motion is reversed, whereas quadratic terms do not (Stark, 1906, 409–10):

$$\frac{\Delta\lambda}{\lambda} \sim \frac{1}{2}\frac{v^2}{c^2}.$$

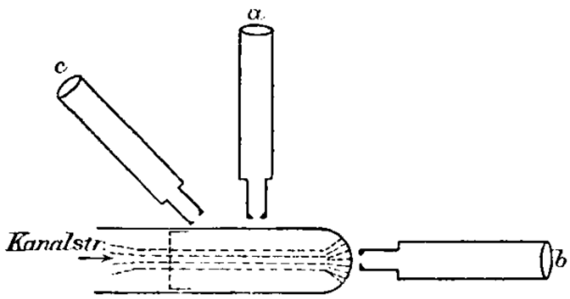

Figure 3: Setup for observing the Doppler shift of canal-ray spectral lines, with spectrographs aligned parallel and perpendicular to the direction of propagation (Stark, 1906, 410)

To test for such a direction-independent effect, Stark arranged observations perpendicular to the direction of ion motion (the spectrograph a in fig. 3). In this transverse configuration, the classical longitudinal Doppler effect vanishes by symmetry, since there is no velocity component along the line of sight.

Stark interpreted any residual shift as potentially dynamical in origin, arising from a velocity-dependent "modification of the electromagnetic forces between the electrons, which is measured by even powers of the ratio v/c" (Stark, 1906, 449). In Stark's model, the ion as a 'carrier' is not a rigid body but a physically deformable system, whose internal state can change with its motion and environment (Stark, 1906, 449). As the translational velocity of the ion increases, the amplitudes of the internal 'centers' of emission (the electrons) were assumed to increase, since the strong electric fields required to accelerate the ions were thought to excite the bound electrons into oscillations of larger amplitude. Moreover, Stark speculated that radiation pressure acting on the oscillating electrons, the finite propagation speed of electromagnetic interactions, or deformation of the oscillatory system at high velocities might contribute to such effects. On this view, the combined action of increased electron amplitudes and radiation pressure could produce a physical deformation of the carrier, which Stark investigated as a possible mechanistic cause of wavelength shifts.

Stark reported tentative empirical indications consistent with this picture, noting a redshift of the midpoints of certain spectral lines with increasing ion velocity—for example, a displacement of about 0.42 Å for the Hγ line at a cathode fall of 7500 Volts (Stark, 1906, 450). He emphasized, however, that while the attainable ion velocities were already sufficient to produce an observable longitudinal Doppler effect, which is first order in v/c, they were still insufficient, in practice, for the transverse effect proportional to v^2/c^2 to be unambiguously isolated. In addition, he stressed the extreme experimental difficulty of ensuring exact transverseness. The slightest radial component would swamp the transverse effect. In particular, he calculated that a misalignment of only $1°59'$ would suffice to generate a spurious redshift of comparable magnitude through the ordinary longitudinal Doppler effect. For these reasons, Stark explicitly characterized these observations as orienting indications rather than as conclusive experimental confirmation (Stark, 1906, 450–51).

At that time, Einstein's new dynamics of free electrons was under attack. The Einstein–Planck approach to electron dynamics was criticized by Ehrenfest (1907): since non-spherical electrons would, in principle, behave differently from spherical ones, the absence of a definite electron model was taken to be a defect rather than a virtue. Einstein (1907a), however, was not particularly impressed by this objection. He famously emphasized that relativity was not a complete theory, but merely imposed kinematical constraints on admissible theories. Einstein did not develop a theory of the electron, but modified the equations of motion of a charged particle in general so as to make them Lorentz invariant. After coming across Stark's (1906) paper toward the end of 1906, Einstein may have quickly realized that this work opened the possibility of a more direct test of the new kinematics itself, without recourse to electron dynamics. He addressed this issue in a paper submitted on March of 1907.

Stark's experiments showed that wave sources moving at very high velocities were in fact available: the positive ions forming canal rays emit line spectra that exhibit a longitudinal first-order Doppler effect, that is, an effect proportional to v/c due to their motion. As we have seen, Stark also attempted to detect an additional second-order effect, proportional to v^2/c^2; however, the experimental setup, not specifically designed for this purpose, was insufficient to yield a reliable result. Nevertheless, it was shown that, in principle, it was possible to decide whether the spectral lines of fast-moving ions exhibit a redshift when they are observed spectroscopically at right angles to the direction of motion. Einstein's aim was to show that the principle of relativity, together with the principle of the constancy of the velocity of light, suffices to account for the effect in question (Einstein, 1907b, 197). It is worth noting, however, that Einstein did not present Stark's work as a test of the relativistic wave equation as a whole, but rather as a test specifically of time dilation, since in the transverse case the effect depends solely on the time transformation.

Referring to his earlier 1905 paper, Einstein reminds his readers that, in relativity theory, a uniformly moving clock runs more slowly, when judged from a 'stationary' system: successive ticks of a clock at rest in K' are assigned a longer coordinate-time interval by the synchronized clocks at rest in a 'stationary' system K than by an observer co-moving with the clock. Einstein appears implicitly to assume that any periodic system passing through identical phases can be regarded as a clock. If ν_0 denotes the frequency defined as the number of periods N divided by the time interval $\Delta t'$ measured in the co-moving system K', and ν denotes the frequency defined as the same number N divided by the coordinate time interval Δt measured in a system K relative to which the atom is moving, then, since $\Delta t = \gamma\,\Delta t'$, special relativity predicts:

$$\frac{\nu}{\nu_0} = \sqrt{1 - \frac{v^2}{c^2}}, \tag{1}$$

Testing this relation requires clocks that can be accelerated to very high velocities without compromising their accuracy. Direct experimental verification using mechanical clocks is therefore likely to remain unattainable, since the velocities that can be imparted to such devices are negligible in comparison with the speed of light. Nature, however, provides systems that effectively function as clocks and that can be accelerated, by means of an electric field, to extremely high speeds: "The atom ion of the canal rays, which emits and absorbs radiation of definite frequencies, is therefore to be regarded as a rapidly moving clock, and the relation just given is thus applicable to it" (Einstein, 1907b, 198).

One might object that the frequency ν_0 of the fast-moving ion cannot be measured by a co-moving spectrograph. However, according to Einstein, there was sufficient evidence to support the claim that, whether the emitting system consisted of neutral hydrogen in a discharge tube or of hydrogen ions produced in canal rays, the same characteristic spectral lines reappeared. This was taken as strong evidence that the Balmer series expresses a stable property of the hydrogen system, rather than something contingent on the manner in which it was produced. Einstein could then point out that the corresponding frequencies are determined solely by the nature of the ion—that is, ions of the same species yield the same frequency ν_0 when measured with a spectrograph at relative rest:

> However, one has to take into consideration that the frequency ν_0 (for the co-moving observer) is unknown, so that the above relation is not accessible to direct experimental verification. But, it may be assumed that ν_0 is also equal to the frequency emitted or absorbed by the same ion while at rest, and this for the following reason. From the fact that one and the same line spectrum is formed under very different conditions, we conclude that the frequency ν_0 does not depend on interactions between moving ions and the stationary gas, but is a characteristic of the ion only; from this one directly concludes with the help of the principle of relativity that ν_0 must equal the frequency of radiation emitted or absorbed by an ion at rest. (Einstein, 1907b, 198)

Einstein then assumes that ions of a given species can be treated as identical systems: the recurrence of the same sharp spectral lines under widely varying experimental conditions supports the assumption that their characteristic frequencies ν_0 are intrinsic properties of the ions themselves and do not depend on interactions with the surrounding medium. That is, all ions of the same species exhibit the same spectral lines when at relative rest. Since canal-ray ions move at very high velocities, any possible shift ν of the spectral lines with respect to ν_0 would thus become, at least in principle, empirically accessible.

The first-order Doppler effect arises from the relative motion between source and observer and depends linearly on v/c. In classical theory it does not presuppose any particular microphysical account of the emission process, nor does it require modifications of the internal dynamics of the emitter. In Stark's work, however, the second-order Doppler effect—that is, any frequency shift proportional to v^2/c^2 observed in a transverse configuration—could not be explained by simple relative motion along the line of sight. Stark therefore treated the effect as a genuinely dynamical modification of the 'carrier': he assumed that uniform motion produces a real change in the emission process of fast ions—a kind of reduction in the number N of oscillations completed by the moving ion during the same universal time interval (frequency contraction). Einstein explicitly rejected this dynamical explanation. Instead, he invoked the principle of relativity to support the assumption that uniform motion does *not* alter the internal oscillatory processes of the ion. The same emission process involves the same count N of completed oscillations; the relativistic effect arises because this same count is referred to different time intervals in the two inertial descriptions.

This assumption is the key to Einstein's reasoning. The intrinsic frequency $\nu_0 = N/\Delta t'$ is taken to be fixed, and longitudinal Doppler components are excluded by definition. As a consequence, the remaining transverse Doppler effect, that is, the reduced observed frequency $\nu = N/\Delta t = \nu_0/\gamma$, can *only* be attributed to the transformation properties of the time coordinate between inertial reference systems in uniform relative motion, $\Delta t = \gamma \Delta t'$, as prescribed by Einstein's new kinematics. The transverse Doppler effect thus serves as a test of the time-dilation relation. Expanding eq. (1) in a series, Einstein (1907b, 197) obtains, to second order in v/c,

$$\frac{\nu - \nu'}{\nu'} \approx -\frac{1}{2}\left(\frac{v}{c}\right)^2 , \tag{2}$$

which quantifies this deficit. Equation (2) therefore provides a direct theoretical expression for the sought second-order effect in the radiation emitted by fast canal-ray ions.

Equation (2) can be tested experimentally by means of a spectrograph oriented such that the motion of the emitting ions is perpendicular to the line of sight; in this configuration, all first-order Doppler contributions vanish. A spectrograph registers the wavelength at which a given spectral line appears after dispersion. The observed position of a spectral line therefore determines its wavelength λ, and, via the relation $\lambda = c/\nu$, its frequency ν. By comparing the wavelength λ of the line emitted by the moving ions with the reference wavelength λ_0 measured for ions at rest, the spectrograph allows one to determine the fractional shift $(\lambda - \lambda_0)/\lambda_0$, which is equivalent (up to a change of sign) to $(\nu - \nu_0)/\nu_0$. Comparing the relativistic prediction with Stark's experimental results, Einstein noted that the numerical values reported by Stark exceeded the expected effect by more than an order of magnitude. However, he expressed skepticism regarding the reliability of the existing measurements (Einstein, 1907b, 198). Nevertheless, in correspondence with Stark in April 1907, Einstein expressed the hope that it might eventually become possible to detect such second-order effects experimentally with greater accuracy (Einstein to Stark, Apr. 13, 1907; Einstein, 1987, Vol. 5, Doc. 45).

2.3 The Transverse Doppler Effect in Einstein's 1907/1908 Jahrbuch Paper

Stark was the editor of the *Jahrbuch der Radioaktivität und Elektronik*. In September, he invited Einstein to contribute a comprehensive review article on the theory of relativity, and Einstein agreed to "deliver the desired report" (Einstein to Stark, Sep. 25, 1907; Einstein, 1987, Vol. 5, Doc. 58). Stark assisted him by supplying relevant literature that the young Einstein could not easily access (Einstein to Stark, Oct. 7, 1907; Einstein, 1987, Vol. 5, Doc. 61). The paper was ultimately submitted on December 4, 1907 and appeared in the *Jahrbuch* at the end of January 1908. Beyond synthesizing his earlier relativity work, Einstein (1908) incorporated several recent developments, including Laue's (1907) derivation of the drag coefficient, Planck's (1906a) reformulation of relativistic dynamics, and related advances in the treatment of thermodynamics within the relativistic framework (Planck, 1907). As expected, Einstein also used the 1908 review to discuss Stark's (1906) observations as a possible means of testing the new kinematics.

A distinctive feature of Einstein's new presentation is that the transverse Doppler effect is no longer treated primarily as a consequence of the Lorentz-invariant wave equation, but is instead relocated within the kinematical framework of the theory. Stark's work on the second-order transverse Doppler effect is mentioned explicitly in the kinematical part I of Einstein's paper, where time dilation is discussed as one of the consequences of the new kinematics (Einstein, 1908, 422). In the electrodynamic part II, by contrast, Einstein derives the Lorentz-invariant wave equation and hence the relativistic Doppler effect without explicitly distinguishing between longitudinal and transverse cases. Although Stark is not mentioned in this context, Einstein does point out that the relativistic Doppler effect can be tested by measuring the frequency shift of light emitted or absorbed by canal rays, which depends both on the velocity of the ions and on the direction in which the radiation is observed (Einstein, 1907a, 425). At the time, canal rays were essentially the only laboratory sources capable of reaching velocities sufficiently large for such effects to be spectroscopically accessible.

In this way, the discussion of the transverse Doppler effect is conceptually shifted from the domain of optics and electrodynamics to that of kinematics (Einstein, 1908, 421–22). The presentation of the time-dilation effect here differs significantly from that of the 1905 paper: it places far less emphasis on algebraic derivation and is instead oriented toward clarifying the physical interpretation and the prospective experimental testability of the effect. Einstein again considers a clock U' permanently at rest at $x' = 0$, at the origin of the moving system K', which moves uniformly with velocity v relative to K along the x-axis. The clock is modeled as a physical periodic system with an intrinsic rest frequency ν_0. The intrinsic frequency ν_0 characterizes the clock as a physical system and is assumed to be independent of its state of uniform motion. Accordingly, the successive ticks of the clock can be labeled by the integers $n = 1, 2, 3, \ldots$, corresponding to the completion of successive periods of the same physical process.

Since the clock U' is at rest in K', that is, since $\Delta x' = 0$, its successive ticks occur at equal intervals of the time coordinate t':

$$t'_n = \frac{n}{\nu_0},$$

where the relation between the discrete tick number n and the continuous time coordinate t' is fixed by the convention that assigns the rate ν_0 to the clock. Accordingly,[4] the time instants at which successive periods are completed in the system K' are

$$t'_1 = \frac{1}{\nu_0}, \quad t'_2 = \frac{2}{\nu_0}, \quad t'_3 = \frac{3}{\nu_0}, \quad \cdots \quad t'_n = \frac{n}{\nu_0}.$$

Einstein then asks how fast this same clock runs when viewed from the system K. To answer this question, he determines the coordinate times in K at which the successive periods of the moving clock are completed. These times are obtained by applying the Lorentz transformation to the time coordinate of K':

$$t = \gamma \left(t' - \frac{v}{c^2} x' \right).$$

[4] In the following I also rely on Einstein, 1910, 131–32 to clarify Einstein's otherwise somewhat elliptical reasoning.

Since the clock U' remains permanently at rest at the origin of K', one always has $x' = 0$. The coordinate time in K corresponding to the completion of the nth period is therefore

$$t_n = \gamma t'_n = \frac{\gamma}{\nu_0}\, n.$$

Accordingly, the successive periods of the moving clock U' are completed, in the system K, at the time instants

$$t_1 = \gamma\,\frac{1}{\nu_0}, \quad t_2 = \gamma\,\frac{2}{\nu_0}, \quad t_3 = \gamma\,\frac{3}{\nu_0}, \quad \cdots \quad t_n = \gamma\,\frac{n}{\nu_0}.$$

Comparing the coordinate-time differences $t_n - t_1$ and $t'_n - t'_1$ assigned to the same first and nth tick of U' in the two systems K and K', one finds that

$$\Delta t = \gamma\,\Delta t'.$$

The frequency of the clock U' at rest in K' is ν_0. When the same clock is described from the system K, the same number N of periods is distributed over a longer elapsed coordinate-time interval $\Delta t = \gamma\,\Delta t'$. From the standpoint of K, U' therefore appears to have a reduced frequency ν:

$$\nu = \frac{N}{\Delta t} = \frac{N}{\gamma\,\Delta t'} = \nu_0\sqrt{1 - \frac{v^2}{c^2}}. \tag{3}$$

In this precise sense, one can say that the moving clock runs ‘more slowly’ than an identical clock at rest in K: the ticks of the moving clock are assigned a longer coordinate-time interval by the synchronized clocks at rest in K than the ticks of an identical clock at rest in K.

The claim that the moving clock runs ‘more slowly’, however, must be treated with some caution. As one can see, Einstein’s argument rests on the fundamental assumption that clocks are *physically identical* across non-accelerated reference frames, that is, that they possess the same intrinsic frequency $\nu_0 = N/\Delta t'$ independently of their uniform motion. Under this assumption, time dilation is attributed to the dilation of the coordinate-time interval Δt assigned in K to the same number N of completed periods of the moving clock. One of the most significant novelties of Einstein’s 1908 paper is that he made this assumption explicit for the first time: “the rate of a clock does not undergo any permanent change if such objects are set in motion and then brought to rest again” (Einstein, 1908, 420; fn. 1). If the clock U' were carefully brought to rest and placed next to any of the clocks U at rest in K, the two would run at the same rate ν_0. This requirement may therefore be taken as a criterion for a ‘good’ clock, namely one whose rate is not permanently affected by its past history.

To test the relation eq. (3) empirically, one requires the existence of a physical system that actually behaves in this way, that is, a real clock capable of reproducing the same time units in different non-accelerated reference frames. *If* such a system exists, *then* the consequences of the new kinematics for time become empirically accessible. As we have seen, nature offers objects that possess the character of ideal clocks, namely ions emitting spectral lines:

The formula $\nu = \nu_0\sqrt{1-\frac{v^2}{c^2}}$ admits of a very interesting application. Mr. J. Stark showed in the previous year that the ions forming canal rays emit line spectra, by observing a displacement of spectral lines that is to be interpreted as a Doppler effect. Since the oscillation process that corresponds to a spectral line is to be considered an intra-atomic process *whose frequency is determined by the ion alone*, we may consider such an ion as a clock of a certain frequency ν_0, which can be determined, for example, by investigating the light emitted by *identically constituted* ions which are at rest relative to the observer. The above consideration shows, then, that the effect of motion on the light frequency that is to be ascertained by the observer is not completely given by the [longitudinal] Doppler effect. The motion also reduces the (*apparent*) proper frequency of the emitting ions in accordance with the relation given above. (Einstein, 1908, 422; my emphasis)

This passage makes clear that the premise of the argument is that ions of the same species are taken to be physically identical when compared at relative rest; they are all 'identically constituted'. By 1908, spectroscopy had established with great precision that atoms and ions of the same species emit sharp, reproducible spectral lines whose frequencies are invariant across laboratories and experimental conditions, provided the emitters are compared at rest. This empirical stability strongly suggested that atomic emission frequencies ν_0 are intrinsic properties of the emitters, not contingent on macroscopic conditions or preparation histories. Despite the lack of a theoretical explanation, for Einstein this made the atom, the "producer of spectral lines" (Einstein, 1908, 459), uniquely well suited to function as a natural clock.

Since ions are assumed to be identical in all frames in uniform relative motion, the frequency shift of fast-moving ions measured by a spectrograph at rest cannot be attributed to a change in the ions themselves. In the longitudinal case, the observed shift may still be explained by motion along the line of sight. By contrast, for a spectrograph oriented perpendicular to the direction of motion, no such explanation is available. In this case, the observed redshift must be understood as a consequence of time dilation alone. For this reason, Einstein described the redshift as *apparent*.[5] The reduction in frequency is frame dependent and is not recorded by a spectrograph co-moving with the ion. The adjective *apparent* was likely chosen to contrast the relativistic account with Stark's dynamical interpretation.

As we have seen in section 2.1, in Stark's view, the motion of the ion produces a *real* frequency contraction, that is, a modification of the emitting system's intrinsic oscillation frequency ν_0, such that fewer oscillations are completed during a given time interval in the moving frame. In Einstein's kinematical interpretation, by contrast, the intrinsic periodic process remains unchanged, since it is fixed by the same physical emission mechanism. The observed redshift is therefore not due to a change in the emitter's frequency ν_0, but to time dilation: relative to a frame K in which the ion is moving, the same completed oscillations are distributed over a *longer* time interval $\Delta t = \gamma \Delta t'$ than in the rest frame K'. The frequency shift is thus explained by the relativistic transformation of the time coordinate, not by a dynamical modification of the source. However, Einstein clearly regarded this 'apparent' effect as an unavoidable consequence of the theory and as one that could, in principle, be empirically detected.

Since the effect is of second order in v/c, it requires clocks moving at sufficiently high velocities, a condition satisfied by ions in canal rays. If two identical spectrographs are at rest with respect to the source but operate for different exposure times, the longer exposure merely collects a larger number N of wave crests: both the elapsed

[5] If a transversely moving source produces a redshift $\nu = \gamma^{-1}\nu_0$, then a transversely moving spectrograph relative to a source at rest produces the inverse effect, a blueshift $\nu = \gamma\nu_0$. The two situations are completely

time and the number of accumulated oscillations increase proportionally. As a result, the inferred frequency ν_0 remains unchanged, and the spectral lines appear at the same position; only the intensity varies. In the relativistic case, by contrast, the crucial point is that the same physical emission process is described from two different non-accelerated frames. The same number of oscillations N associated with a given emission process is thus referred to different elapsed times, $\Delta t = \gamma \Delta t'$. The difference in the temporal interval over which the same oscillations N are distributed necessarily entails a change in the observed frequency ν. Through the relation $\lambda = c/\nu$, this difference in temporal rate is translated into a difference in wavelength, which the spectrograph at rest in K registers directly as a redshift of the atom's spectral line.

3 Ritz and the Transverse Doppler Effect as an *Experimentum Crucis*

Replying to a letter from Sommerfeld that is no longer extant, Einstein expresses strong interest in having a canal-ray experiment carried out and welcomes Sommerfeld's willingness to involve Peter Paul Koch (at that time an assistant of Wilhelm Röntgen in Munich) in testing the transverse Doppler effect. Einstein nevertheless betrays some caution as to whether Koch would actually carry out the experiments (Einstein to Sommerfeld, Jan. 14, 1908; Einstein, 1987, Vol. 5, Doc. 73). He notes that Stark had already informed him some months earlier of his intention to undertake the same investigation, although Stark had since written to Einstein repeatedly without again mentioning the experiment (Einstein to Sommerfeld, Jan. 14, 1908; Einstein, 1987, Vol. 5, Doc. 73). The letter to Sommerfeld also shows that this was something of a side issue. The main point of debate concerned the empirical test of the variability of the electron mass investigated by Kaufmann. Indeed, Einstein's 1908 paper devotes the entire §10 to a discussion of Kaufmann's experiments and to a comparison between the different predictions of the Abraham electron, the Einstein–Lorentz electron, and the Bucherer electron (Walter, 2018).

These electron models were all constructed within a theoretical context shaped by the assumed validity of Maxwellian electrodynamics and by constraints imposed by optical and electrodynamic phenomena, even though they differed markedly in their interpretations of the status and significance of the Lorentz transformations. A more radical challenge to relativistic kinematics was posed by Ritz, who in the winter of 1907–1908 moved from Göttingen to Tübingen, where he worked with the experimentalist Friedrich Paschen. Regarded as a rising star of European physics for his work in spectral theory (Hentschel, 2012), Ritz became increasingly skeptical of Lorentz–Maxwell electrodynamics after a stay in Leiden with his close friend Ehrenfest (Martinez, 2004). His major polemical paper appeared in the *Annales de chimie et de physique* in February 1908, only a few weeks after the publication of Einstein's 1908 review article. Like Einstein, Ritz (1908a, 147–48) was skeptical of the reliability of Kaufmann's experiments. However, for him, the difficulties involved in constructing a suitable model of the electron and in accounting for the velocity dependence of its

symmetrical in principle, but only the redshift from a transversely moving source is usually referred to as the transverse Doppler effect. While fast-moving atoms are readily available in canal rays, fast-moving spectrographs are not experimentally feasible. Nevertheless, this symmetry shows that the transverse Doppler effect cannot be attributed to a real, direction-dependent dynamical deformation of the atom.

mass were not isolated technical problems but rather symptoms of a deeper flaw in Maxwell–Lorentz electrodynamics itself.

Lorentz–Maxwell electrodynamics assumes the existence of an absolute system of ether coordinates, taken to be independent of the motions of matter, with respect to which light moves with velocity c. In order to reconcile this framework with the failure of the Michelson–Morley and other ether-drift experiments, it became necessary to eliminate the absolute system by means of increasingly implausible auxiliary hypotheses (Ritz, 1908a, 148). As Ritz complained, this strategy, pursued consistently by Lorentz and Einstein, required modifying the principles of kinematics themselves: abandoning absolute time and replacing 'rigid' with 'deformable' bodies, while also treating the parallelogram rule for the composition of velocities as a mere first approximation, valid only at low speeds. However, "[i]t would be regrettable, for the economy of our thinking, if we were forced to admit such complications" (Ritz, 1908a, 148). By abandoning Maxwellian electrodynamics in favor of a description based solely on retarded potentials, Ritz was led naturally to a ballistic theory of light, in which radiation propagates with the velocity of its source, $c \pm v$.

Ritz (1908c, 203) was well aware that the approaches of Lorentz and Einstein were not identical. Lorentz was committed to the reality (*réalité*) of length contraction and time dilation, whereas for Einstein lengths and times only appear (*semblera*) to be contracted and dilated, while true lengths and times remain invariable (Ritz, 1908c, 203). Nevertheless, this difference appeared to him to be marginal, since both approaches led to the same observable consequences (Ritz, 1908b, 213). After returning to Göttingen in the spring of 1908, Ritz, in correspondence with Paschen and as a leading expert in spectroscopy, naturally singled out the transverse Doppler effect in canal rays as an experimentally decisive test case capable of discriminating between classical kinematics and the new kinematics:

> When I was in Tübingen, you had a hydrogen tube in operation for canal-ray investigations. I would like to present to you a problem which is of the greatest importance for the question of the principle of relativity, and thus for the whole of electrodynamics. According to the Lorentz–Einstein theory of relativity, the wavelength emitted by a moving atom must change not only in the direction of motion in accordance with the Doppler principle; when observed perpendicular to the direction of the velocity, there must also occur a red shift of magnitude $\frac{1}{2}\left(\frac{p}{c}\right)^2\lambda$ (c = speed of light). For H_γ and $v = 1000$ km/sec, this would amount to about 0.02 Å.
>
> In observations of canal rays, one would therefore observe an apparent shift of the line by less than 0.02 Å, since the stationary and moving intensities could not, of course, be separated. With fine lines, however, and if one had the normal and the shifted spectrum on the same plate, this displacement could be detected even without a large grating. Would it not be possible to arrange matters so that the question of the existence of this effect could be answered with certainty?
>
> If the effect exists, then it is the end of our universal time, of the parallelogram of velocities, and of the whole of kinematics. I hope the effect does not exist—that would be much nicer. But since this is a genuine *experimentum crucis*, and since you may perhaps, despite the great difficulty, find a way to realize the matter, I wished to communicate it to you. (Ritz to Paschen, Jun. 14, 1908; Ritz, 1911b, 523–24)

The null result of the Michelson–Morley experiment could be explained equally well by assuming a ballistic character of light emission. The transverse Doppler effect, by contrast, cannot be accommodated within such an emission theory. Thus, Ritz was the first to characterize explicitly the transverse Doppler effect as an *experimentum crucis* for discriminating between his emission theory and Lorentz–Einstein relativistic kinematics itself, independently of more indirect consequences of the latter, such as the velocity dependence of the electron's mass. In all non-relativistic theories, including emission theories, a longitudinal Doppler effect can be accounted for, even though the underlying explanations differ. By contrast, only Lorentz–Einstein relativity predicts a genuinely relativistic Doppler component transverse to the direction of motion.[6]

After the meeting of the German Association of Natural Scientists and Physicians in Cologne in September 1908, many physicists felt compelled to favor relativity, driven both by Minkowski's (1909) four-dimensional reformulation of the theory and by Bucherer's (1908) alleged experimental confirmation of the Lorentz–Einstein electron. However, Ritz's subsequent correspondence with his friend Ehrenfest suggests that he remained unconvinced (Staley, 2008, 268–271): "Minkowski forcefully promotes the principle of relativity *à la* Einstein. Hopefully this nonsense will disappear in 10 years. A wholly miserable makeshift [*Auskunfts-mittel*]. One could salvage any theory, no matter how false, in such a way" (Ritz to Ehrenfest, Dec. 17, 1908, cit. in Staley, 2008, 270f.; translation modified). Rather than tinkering with the classical kinematical framework, Ritz (1908c) believed that many of the difficulties of contemporary physics, including the so-called ultraviolet catastrophe, could be resolved by reconstructing the foundations of electrodynamics through the abandonment of the ether and the exclusive use of retarded potentials.

Einstein (1909b) was the first to offer a systematic reply to this line of criticism, prompting Ritz's (1909) rejoinder; Ritz and Einstein (1909) ultimately agreed to disagree. Ritz maintained that irreversibility (the exclusion of the advanced potentials) must be imposed at the level of the fundamental electrodynamic laws, whereas Einstein held that it emerges only at the statistical level. In his March 1909 habilitation lecture on the principle of relativity in optics, Ritz (1911a) reiterated the stark dilemma facing physics at that time: (a) to preserve Maxwell electrodynamics and the wave theory of light with a source-independent velocity c at the price of adopting a new kinematics, or (b) to preserve the classical kinematical framework by radically modifying electrodynamics in the direction of an emission theory of light, in which fictitious light corpuscles would travel with velocities $c \pm v$. Two months after Ritz's death, in September 1909, Einstein, in his Salzburg lecture on the constitution of radiation (Einstein, 1909a), outlined, without mentioning Ritz, a contrasting program. Whereas Ritz sought to preserve classical kinematics and the continuity of radiation by abandoning the constancy of the speed of light, Einstein instead accepted the granular character of radiation and upheld the universal constancy of the speed of light by introducing a new kinematics.

4 Real and Apparent. Einstein's Last Remarks on the Transverse Doppler Effect

While struggling with the problem of radiation toward the end of 1909, Einstein (1910) wrote a review paper on relativity, published in French in two installments in the *Archives des sciences physiques et naturelles*. By his own admission, the paper contained nothing physically new (Einstein to Laub, Aug. 27, 1910; Einstein, 1987, Vol.

[6]Although the transverse Doppler factor can be retroactively read into Lorentz's equations, Lorentz never identified, interpreted, or proposed it as a physical effect. The effect is not mentioned in Lorentz's (1916) 1909 Columbia lectures.

5, Doc. 224). Nevertheless, at least one conceptual novelty can be identified. Einstein explicitly addresses the question "What is a clock?" (Einstein, 1910, 21). Einstein famously defined a clock as any periodic system completely isolated from external influence that repeatedly passes through identical phases, such that—by the principle of sufficient reason—each period is taken to be identical to any other: "We therefore postulate that two identical phenomena have the same duration. The perfect clock thus defined plays, for the measurement of time, a role analogous to that of the perfect rigid body in the measurement of lengths" (Einstein, 1910, 21; fn. 1). After identifying the beginning and the end of a process, one can count the number N of identical cycles that occur. If the clock takes the form of a mechanism equipped with hands, completed cycles can be labeled by the integers $n = 1, 2, 3, \ldots$ via the positions of the hands. Assuming a stable intrinsic frequency of the clock ν_0 as a unit of time, this counting perspective can be translated into a measuring perspective. A clock at rest in a co-moving system directly measures the time coordinate of that system, such that the time coordinate of the nth tick is given by $t'_n = n/\nu_0$.

One may then raise the separate question of whether systems exist in nature that realize the ideal of a perfect clock, that is, systems that ensure the physical stability of the frequency and therefore of the time unit. In classical physics, two objects could never be identical in every respect, since a complete physical description would, in principle, require infinitely many parameters, allowing systems to differ in arbitrarily small details. This is precisely why Einstein's appeal to atomic or ionic processes in the section dedicated to time dilation is epistemologically significant (Pierseaux, 2003). Spectroscopy provides strong empirical evidence that atoms and ions of the same species exhibit the same characteristic frequencies when compared at rest, regardless of how they were previously accelerated. Thus, these systems realize, in nature and to a high degree of accuracy, Einstein's definition of perfect clocks: "Since the oscillatory phenomena that give rise to a spectral line must be regarded as intra-atomic phenomena whose frequency $[\nu_0]$ is determined solely by the nature of the ions, we may take these ions to serve as clocks" (Einstein, 1910, 132–33).

The proper frequency ν_0 of the oscillatory motion of the ions provides a means of measuring time t: "this frequency will be known if one observes the spectrum produced by ions of the same nature, but at rest with respect to the observer" (Einstein, 1910, 133). Ions can be accelerated in canal rays by electric fields. Einstein clearly believed that there was sufficient empirical evidence to justify the assumption that acceleration does not permanently affect ions, which were expected to exhibit the same spectral lines when measured by a co-moving spectrograph. If this assumption is granted, the theory predicts that "there is an influence of the motion on the source, which reduces the *apparent* [*apparente*] frequency of the ion" (Einstein, 1910, 133; my emphasis), as measured by a spectrograph when the motion of the source is perpendicular to the line of sight.

As suggested above, the use of the term *apparent* is likely intended to avoid confusion with Stark's interpretation of this 'influence' as a *real* dynamical contraction of the atomic frequency ν_0, that is, a reduction in the number of periods completed during the same time interval in the same frame. In Einstein's reading, by contrast, clocks retain the same intrinsic frequency ν_0. Since there is no relative motion along the line of sight, the observed frequency shift $\nu = \gamma^{-1}\nu_0$ can only be due to the fact that the same number N of periods is associated with different time intervals in different inertial frames:

$$\nu = \frac{N}{\Delta t} = \frac{N}{\gamma \Delta t'} = \frac{\nu_0}{\gamma} = \nu_0 \sqrt{1 - \frac{v^2}{c^2}}.$$

Thus, the Lorentz factor enters the expression not because of a 'real' reduction of the emitter's intrinsic frequency ν_0. The frequency shift $\nu = \gamma^{-1}\nu_0$ arises because, in a non-co-moving frame K, the same completed oscillations are referred to a longer coordinate time interval, $\Delta t = \gamma \Delta t'$. The oscillations are therefore less densely packed in time in the non-co-moving frame, which manifests itself as a redshift without any change in the intrinsic emission process given by $\nu_0 = N/\Delta t'$. In this sense, the effect is 'apparent': the shifted line is not detected by a spectrograph at relative rest with respect to the moving atom, which records the proper frequency instead.

In the ensuing months, following the debate raised by Ehrenfest's paradox, Einstein realized that the terms 'apparent' and 'real' gave rise to misunderstandings (Einstein to Varićak, Feb. 24, 1911; Einstein, 1987, Vol. 5[10], Doc. 255a). Some regarded kinematical effects, such as 'time dilation', as mere coordinate effects and therefore as 'apparent'. By contrast, Stark's 'frequency contraction' was taken to be 'real' as a dynamical effect. To avoid confusion, Einstein suggested disentangling the conceptual pair real/apparent from kinematical/dynamical (Einstein, 1911). Relativistic time dilation, although a kinematical effect, is both apparent and real. It is 'apparent', since a co-moving clock or spectrograph records the corresponding proper quantity; yet it is *real*, since it can be tested empirically. Given a suitable transverse observational arrangement, any spectrograph in relative motion with respect to the emitting ion will measure a shifted line; a spectrograph co-moving with the ion will measure the proper line. The 'apparent' change of frequency is nevertheless 'real' in the sense that it would be absent in a rival theory.[7] As Ehrenfest pointed out, this is precisely the reason why, in his letter to Paschen, his friend Ritz[8] proposed the transverse Doppler effect as an *experimentum crucis* to decide between classical and relativistic kinematics[9]: "Whereas Einstein's theory *unconditionally* demands the existence of this effect [...][,] Ritz's emission theory *unconditionally* demands its nonexistence" (Ehrenfest, 1912, 319; my emphasis).

Conclusion

As is well known, Einstein repeatedly insisted that the new kinematics is not merely a matter of the conventional synchronization of clocks. Once established, the Lorentz transformations are either true or false in the sense that they can be *confirmed* or

[7]This case is by no means an *unicum* in the history of physics. A useful comparison can be drawn with stellar aberration in classical ether theory, which is likewise both *apparent* and *real*. It is apparent insofar as it disappears for a suitably chosen observer, yet real insofar as it is empirically detectable. Over the course of the year, astronomers must therefore inevitably reorient their telescopes toward the *apparent* position of the stars, whose annual displacement traces ellipses, although, of course, their *real* positions have remained unchanged. Nevertheless, stellar aberration is undeniably a genuine physical effect that would not be expected in a theory of complete ether drag..

[8]The letter was later published in the 1911b edition of Ritz's collected works.

[9]In the absence of a direct confirmation of the light postulate, which would soon be provided by de Sitter (1913).

disconfirmed by their empirically accessible consequences, provided that the coordinates are interpreted as quantities measurable by means of 'good' rods and clocks. If atoms, the emitters of spectral lines, are such good clocks that they reliably register the time coordinate in their co-moving frame, then the transverse Doppler effect is indeed one such consequence. At the time, it was the only one that could feasibly be detected. Einstein's line of argument, disentangled from the historical details, may be reconstructed as follows:

1. *Empirical premise (spectral identity of atomic clocks at rest).* Atoms (or ions) of the same species, when compared at relative rest, exhibit the same intrinsic frequency ν_0, as evidenced by the reproducibility of their spectral lines.
2. *Physical premise (spectral identity of atomic clocks in uniform motion).* Uniform translational motion does not alter the intrinsic oscillatory process responsible for emission. Hence identical ions, when compared at rest, possess the same intrinsic frequency $\nu_0 = N/\Delta t'$, independently of their state of uniform motion[10]
3. *Experimental restriction (transverse configuration).* The observation is arranged so as to eliminate longitudinal Doppler components, that is, any contribution due to motion along the line of sight.
4. *Prediction.* A Doppler shift in the frequency $\nu_0 = N/\Delta t'$ (for an atom at rest in K') persists even in the transverse case, namely

$$\nu = \gamma^{-1}\nu_0.$$

5. *Explanation.* The same emission process involves the same count N of completed oscillations. What changes is not this count, but the time interval with respect to which it is evaluated: the same N is referred to different time intervals in the two inertial descriptions:

$$\Delta t = \gamma\, \Delta t'.$$

In this form, the argument amounts at most to a provisional compromise. Whereas premise 3 is a definitional restriction built into the experimental arrangement, Einstein did not yet possess a genuine theoretical account of why atoms or ions possess stable, motion-independent proper frequencies in accordance with premises 1 and 2. At this stage, he could rely only on the overwhelming empirical evidence provided by spectroscopy. It is well known that, beginning in the 1920s, Einstein increasingly acknowledged that a dynamical explanation of the periodic processes used as clocks would ultimately be required (Einstein, 1921, 1923, 1924, 1926). From a more fundamental standpoint, atomic clocks should be regarded as particular solutions of the underlying dynamical equations, with their behavior fixed by invariant parameters of the theory. Such parameters determine characteristic frequencies,[11] which in turn set the intrinsic periodicity of the admissible solutions. If the fundamental theory is

[10] This premise corresponds to what is often called *boostability* (Brown, 2005, 30, 81, 121). As discussed above, Einstein made this assumption explicit as a criterion for 'good' rods and clocks in Einstein, 1908, 420; fn. 1 and 126; fn. 2 Einstein, 1910..

[11] For example, a mass parameter fixes the Compton frequency $\nu_C = mc^2/h$..

Lorentz invariant, the same periodic solutions must occur in all inertial systems, with the same proper frequency ν_0, whether atomic or otherwise.

If such a theory were available, premises 1 and 2 would no longer be merely assumed but would be derived as consequences of the theory itself. Such a theory would thus provide a dynamical explanation of the spectral identity of atoms—the empirical fact that identical atoms, when compared at relative rest, possess the same intrinsic frequency $\nu_0 = N/\Delta t'$ as measured in the co-moving frame K'. Since premise 3 is merely definitional, there would then be no way to avoid conclusion 5, namely that prediction 4, the transverse Doppler effect, is an *apparent* coordinate effect:

$$\nu = \frac{N}{\gamma \Delta t'} = \frac{\nu_0}{\gamma} = \nu_0 \sqrt{1 - \frac{v^2}{c^2}}. \tag{4}$$

The transverse Doppler effect appears because the same number N of completed oscillations of the same process is spread over a longer time $\Delta t = \gamma \Delta t'$ in the non-co-moving system K. Thus, somewhat paradoxically, once a genuine dynamical explanation for the behavior of clocks is established, one must concede that clock retardation admits only a purely *kinematical explanation*. Frequency is nothing more than a ratio. If the numerator is held fixed, the observed frequency shift must be attributed to the denominator.

The only logically possible alternative is to hold the denominator fixed and attribute the same shift to the numerator instead. This would amount to denying premises 1 and 2 and, as Stark envisaged, searching for a dynamical explanation of the *real* contraction of the intrinsic frequency ν_0:

$$\nu = \frac{\gamma^{-1} N}{\Delta T} = \frac{\nu_0}{\gamma} = \nu_0 \sqrt{1 - \frac{v^2}{c^2}}. \tag{5}$$

When the atom moves through the ether, the electromagnetic forces binding it are modified, so that the atom genuinely oscillates more slowly in the co-moving system K': the reduced number $\gamma^{-1}N$ of completed oscillations is spread over the same universal time ΔT. One thereby obtains a *dynamical explanation* of the observed frequency shift, but at the price of postulating an undetectable privileged frame K in which atoms oscillate at their natural frequency ν_0. Somewhat ironically, this approach was pursued by Herbert E. Ives (Ives and Stilwell, 1938) in connection with the experimental confirmation of the transverse Doppler effect in the late 1930s (Lalli, 2013). Ives makes no mention of special relativity; instead, he presents his result as evidence that the change in frequency "predicted by the Larmor-Lorentz theory is verified" (Ives and Stilwell, 1938, 225; see Ives and Stilwell, 1941, 374). Unsurprisingly, Einstein responded rather tersely to Ives' announcement.[12]

In the ensuing years, Einstein continued to concede that it was inconsistent to treat clocks "as independent physical objects" (Einstein to Swann, Jan. 24, 1942; Einstein Archives, n.d., 20-624) as he did in 1905. However, in confessing his youthful 'sin', Einstein (1949a, 61) was clearly not calling for a dynamical explanation of time dilation *à la* Ives (1947). Rather, his concern was the problem of its *confirmation*. The

[12] Ives to Einstein, Jun. 24, 1938; Einstein Archives, n.d., 53-549; Einstein to Ives, Aug. 4, 1938; Einstein Archives, n.d., 80-269.

transverse Doppler effect confirms the Lorentz transformation of the time coordinate only *if* atoms are assumed to be 'good' clocks that, once placed at rest in an inertial frame under identical physical conditions, always tick at the same rate, *i.e.*, emit the same spectral lines. However, special relativity cannot prove whether this is the case. Strictly speaking, clocks should emerge as "solutions of the basic equations" of some fundamental theory (Einstein, 1949a, 59). Yet if such a theory were available, it would be necessary to reconsider what it means to provide 'empirical confirmation' of a prediction of the Lorentz transformations. The latter would then be tested using clocks that are themselves solutions of a Lorentz-invariant dynamical theory. Relativistic kinematics could no longer be confronted with experience independently of Lorentz-invariant dynamics. Rather, only the empirical success of the latter could provide indirect evidence for the Lorentz transformations.

From the standpoint of confirmation, kinematics and dynamics would not confront experience separately, but rather, as Einstein frequently noted, only as a whole that "in its entirety validates itself" (Einstein, 1949b, 678). From the standpoint of *explanation*, however, the separation between kinematics and dynamics within that whole remains intact. Lorentz-invariant laws of matter provide a *dynamical* explanation of why all atomic clocks of the same kind possess the same proper frequency ν_0 in all inertial systems. As we have seen, once this premise is established, the observed transverse shift of spectral lines admits *only* a purely *kinematical* explanation as a time-coordinate effect. Any further dynamical explanation within special relativity would lack an independent explanandum. The case of the transverse Doppler effect shows with particular clarity that, from a historical point of view, Einstein's call for a dynamical explanation of the behavior of clocks was not intended to be a call for a dynamical explanation of time dilation. From a theoretical point of view, however, the pursuit of such an explanation within special relativity can, of course, proceed on a different basis, without appealing to Einstein's authority.[13]

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[13]As is well known, John S. Bell (1976) suggested a third alternative between Einstein and Larmor dilation. One can in principle recover a relativistic effect such as the transverse Doppler effect by calculating how the physical forces within a moving atom are modified on the basis of a Lorentz-covariant theory formulated relative to a single chosen *inertial* frame, namely, the one in which the spectrograph is at rest. Bell's *pedagogical* intent was to provide an effective *illustration* that relativistic effects can be reconstructed dynamically within any chosen inertial frame. In the *philosophical* literature, it has been argued that Bell's proposal amounts to a genuine dynamical *explanation* of such effects (Brown and Pooley, 2001; Brown, 2005, sec. 1.4). However, it is important to note what this approach would entail. Since every inertial frame treats its co-moving atoms as oscillating at the proper frequency ν_0, Bell's proposal generates not one but infinitely many different equally legitimate, frame-relative dynamical explanations of one and the same observed frequency shift ν. The 'pre-established harmony' among these explanations ultimately rests on Bell's assumption that *all* fundamental interactions are Lorentz invariant (Bell, 1976, 143). The crucial question in the dynamical/kinematical debate is ultimately how this '*all*' is to be understood: as a remarkable coincidence among otherwise independent available dynamical laws, or as a requirement imposed on any admissible dynamical law (Lange, 2016). Whether this requirement is expressed algebraically (Einstein) or geometrically (Minkowski) seems to me to be ultimately of secondary importance.

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Is Time Dilation 'Real'? Einstein and the Transverse Doppler Effect

Abstract

As early as 1907, Einstein realized that the second-order transverse Doppler redshift reported by Stark in the spectral lines of fast-moving ions in canal rays could serve as a direct experimental test of time dilation. He rejected Stark's *dynamical explanation* of the effect as a real frequency change of the moving ions. Instead, assuming that ions of the same kind emit the same spectral lines when at relative rest, Einstein offered a *kinematical explanation* of the frequency shift as a consequence of time dilation. He labeled the effect *apparent*, since the shifted line is not recorded by a spectrograph co-moving with the ion. At the same time, he regarded it as *real*, in the sense that it would not occur if the old kinematics held, and is thus empirically testable. From the 1920s onward, Einstein argued that the behavior of atomic clocks calls for a dynamical explanation. The present paper shows, however, that even if a special relativistic explanation of the spectral identity of atoms were available, it would merely reinforce the claim that relativistic time dilation admits a purely kinematical interpretation. It concludes that modern dynamicists misread Einstein's plea for a dynamical account of clock behavior as a demand for an *explanation* of time dilation, whereas it was primarily motivated by the problem of its *confirmation*.

Keywords: Einstein, Special Relativity, Time Dilation, Transverse Doppler Effect, Confirmation, Explanation

Introduction

Over the last two decades, philosophical discussions of special relativity have repeatedly returned to the question of whether relativistic effects, most notably length contraction and time dilation, ultimately call for a *dynamical* explanation (Brown, 2005; Brown and Pooley, 2006), a *kinematical* explanation (Norton, 2008; Janssen, 2009), or whether one should move beyond this binary choice altogether (Acuña, 2016). While the debate is primarily theoretical in intent (Read, 2020; Acuña and Read, 2026), proponents of the dynamical approach have often framed it against a historical background. In particular, it has been argued that, although Einstein initially presented length contraction and time dilation as merely *apparent* coordinate effects, he was ultimately aiming at an account in which these effects reflect *real* physical changes in the equilibrium configurations of moving atomic systems (Brown and Read, 2022). Indeed, he later acknowledged that rods and clocks should have been treated as solutions of the underlying dynamical laws (Einstein, 1949a, 61). It has therefore been suggested that Einstein himself may, at least retrospectively, be seen as gesturing toward a dynamical approach (Brown and Read, 2022, 70).

This paper aims to further challenge this historical narrative, in the hope that this result will bear directly on contemporary debates concerning the theoretical role of the ~~dynamical-kinematical~~ dynamical–kinematical distinction. As recent literature has shown (Giovanelli, 2023), concerns about the proper use of the categories 'real' and 'apparent', as well as 'kinematical' and 'dynamical', had already emerged in the early 1910s. ~~Starting from Max Born's (1909) definition of rigid motion, Paul Ehrenfest (1909) formulated his famous paradox, arguing that a Born rigid disk could not be set into rotation without the emergence of stresses, thereby revealing a contradiction within relativity theory. Some~~ Reacting to Ehrenfest's (1909) paradox, some contemporary relativists (Ignatowski, 1910; Varićak, 1911) ~~challenged Ehrenfest's conclusion. They argued that such stresses~~ argued that stresses in a rotating disk would arise only if 'Lorentz contraction' were interpreted as a *real* dynamical effect, produced by molecular forces within the material of the rod. By contrast, 'Einstein contraction' was said to be merely an *apparent* coordinate effect resulting from an arbitrary synchronization of clocks. It could therefore be made to disappear by adopting a different synchronization convention. ~~On this basis, they concluded that no stresses would arise in a rotating rigid disk~~Relativistic length contraction, Varićak argued, "is only a psychological and not a physical fact, *i.e.*, the body has not really undergone any change" (Varićak, 1911, 169).

~~Both Ehrenfest (1910) and Einstein (1911) , however, felt the need~~ Einstein (1911) felt compelled to take a public stance against this view. Length contraction is indeed *apparent* insofar as it is ~~a coordinate effect that vanishes for a~~ absent for observers co-moving ~~observer, but~~ with the rod; yet it is nevertheless *real* ~~, since it inevitably appears for non-co-moving observers and therefore cannot be eliminated in all frames at once. For this reason, it gives rise to physically meaningful and,~~ in ~~principle, empirically testable consequences: (Einstein, 1911, 510). As Born (1909) showed, in the case of linear acceleration the proper length of a rod can, in principle, be preserved by appropriately coordinating the accelerations of its different parts. In the case of a rotating disk, by contrast, no global~~ Einstein's own sense, namely, measurable by means of rods and clocks. Consider a rod AB at rest in K' and moving relative to K. Its *projection* in K is the segment $A'B'$ determined by the positions A' and B' of the endpoints A and B, measured simultaneously in K. Relativity theory predicts that $A'B' < AB$. In principle, this prediction could be confirmed or disconfirmed by measuring AB and $A'B'$ with the same unit rod, first while it is momentarily co-moving ~~inertial frame exists, so Lorentz contractions along the rim cannot be eliminated simultaneously. The resulting mismatch between circumference and radius gives rise to internal stresses, illustrating how coordinate effects can have genuinely physical consequences~~[1] with AB and then after it has been brought to rest

[1] ~~Note that this situation is closely analogous to Bell's (1976) rocket thought experiment, which is commonly taken to support a dynamical rather than a purely kinematical reading of relativistic length contraction (Brown and Pooley, 2001). The claim is that, if length contraction were a merely innocuous , the emergence of stresses and the possible structural failure of the rope (or of the rotating disk) would be difficult to account for. By contrast, Einstein used the to show that a effect can nevertheless be physically , as showb by the stresses in 's disk~~.

in K without any alteration of its length. As Einstein put it to Varićak: "The contraction can be ascertained by measurement, *i.e.*, it is 'real'" (Einstein to Varićak, Mar. 3, 1911; Einstein, 1987, Vol. 5[10], Doc. 257a).

This paper argues that the same dialectic between the *apparent* and the *real* emerges ~~even~~ more clearly in the case of time dilation. Like length contraction, time dilation is 'apparent' ~~, since it is a coordinate effect that vanishes for a suitably chosen co-moving observer~~insofar as a clock at rest in its own frame records its proper rate; yet it is 'real' ~~, since it cannot be removed for all non-co-moving observers simultaneously, provided that the new kinematics hold~~insofar as it has empirically testable consequences.[1] In ~~this case, no comparable philosophical controversy arose, largely because there was no genuine Lorentzian counterpart to time dilation (Rindler, 1970). Nevertheless, in~~ contrast to length contraction, however, the empirical testability of time dilation soon appeared to be ~~reasonably~~ practically feasible. As early as 1907, Einstein appealed to Johannes Stark (1906)'s experimental work on fast-moving ions in canal rays, which seemed to suggest the presence of a second-order Doppler effect in the transverse direction of motion, that is, even ~~in the case in which the classical~~ when the classical longitudinal Doppler effect disappears. A concrete possibility for testing relativistic kinematics was thus at hand (Giuliani, 2013).

As the paper will show, from the outset, Einstein (1907b) rejected Stark's (1906) original explanation, according to which ions in motion were thought to undergo a 'real', dynamical contraction of their intrinsic frequency. Spectroscopy showed that atoms and ions of the same species always display the same characteristic spectral lines when compared at rest, independently of their prior acceleration history. The principle of relativity implies that atoms of the same kind realize the same intrinsic frequencies in all inertial systems. In the transverse Doppler configuration, there is no relative motion along the line of sight, and hence no classical Doppler contribution. Einstein could then conclude that the reduced frequency must therefore be ascribed to the relativistic relation between time coordinates in different inertial frames, that is, to time dilation itself. Accordingly, Einstein initially labeled the change in the observed frequency 'apparent' in order to emphasize that the intrinsic frequency of ions is supposed to be invariant (~~Einstein, 1908, 422; Einstein, 1910, 133~~Einstein, 1908, 422; Einstein, 1910, 133). Nevertheless, the transverse Doppler effect is 'real' in Einstein's sense, since, ~~since the frequency shift necessarily appears for non-co-moving observers~~ if the new kinematics ~~hold~~holds, a non-co-moving spectrograph would measure the shifted spectral lines, whereas a co-moving spectrograph would measure the ion's proper frequency. The transverse Doppler effect thus offered, at least in principle, a direct test of time dilation, provided that atomic spectral emitters can serve as 'good' clocks.

From the 1920s onward, Einstein explicitly emphasized that the behavior of atomic clocks could not yet be derived from a complete microscopic theory (Einstein, 1921, 1923, 1924, 1926, 1949a). He conceded that, in a fully developed theory, the existence and behavior of clocks would, in principle, emerge as consequences of the fundamental equations themselves. Any theory capable of supplying oscillatory processes usable

[1] In what follows, I use the term 'kinematics' and avoid explicit reference to the 'geometry' of spacetime, since Minkowski's framework was far from being widely adopted in the period discussed in this paper.

as clocks would have to single out solutions characterized by a definite, invariant period, determined by the structural constants of the theory. If the theory were Lorentz invariant, such solutions would recur in all inertial frames in the same way, thereby guaranteeing the identical behavior of clocks when compared at rest. From a modern perspective, an example of such an invariant scale is provided by a mass parameter in relativistic quantum theory, which fixes a characteristic frequency, the Compton frequency $\nu_C = mc^2/h$.

Such a theory would then indeed provide a *dynamical explanation* of why all atomic clocks are *identical* when compared in the same rest frame and therefore emit the same spectral lines. However, this line of reasoning leads to an inevitable conclusion. Given the spectral identity of atoms, and assuming that there is no contribution from motion along the line of sight, the transverse frequency shift must inevitably admit a purely *kinematical explanation*, namely as a consequence of time dilation itself. Somewhat paradoxically, the full realization of the dynamical program shows that time-dilation effects, including the transverse frequency shift, are purely kinematical. In this sense, this paper suggests that the transverse Doppler effect offers a particularly illuminating case for reassessing the long-standing debate between kinematical and dynamical interpretations of special relativity.

In particular, the paper concludes that much of the misunderstanding arises from the fact that Einstein's demand for a dynamical account of rods and clocks has been read as a demand for an *explanation* of the new kinematics, whereas it was primarily motivated by the problem of its *confirmation*. Einstein never tired of emphasizing that relativistic kinematics was not merely conventional but, in principle, testable by means of rods and clocks. Yet rods and clocks are complex physical systems governed by dynamical laws that are themselves supposed to conform to relativistic kinematics. From a logical point of view, kinematics and dynamics can receive empirical confirmation only as a whole. Nevertheless, within this whole, dynamical and kinematical explanations remain distinct. The paper argues that the transverse Doppler effect brings this out in especially vivid terms: dynamics explains the sameness of clocks' frequencies when compared at rest, while kinematics alone explains the frequency shift across inertial frames.

1 Time Dilation and the Transverse Doppler Effect in Einstein's 1905 Relativity Paper

Although Einstein's 1905b relativity paper is among the most cited in the history of science, a brief review of its basic structure may nevertheless be useful for locating the respective derivations of time dilation and the transverse Doppler effect. Starting from the two postulates—the relativity principle and the light postulate—the first, kinematical part of the paper aims to develop a new kinematics of "rigid bodies (coordinate systems), clocks, and electromagnetic processes" (Einstein, 1905b, 892) that resolves their apparent incompatibility. In §1, by introducing the synchronization of clocks using light rays, Einstein shows that there is no *a priori* reason to assume that $t' = t$, that is, that two events which are simultaneous with respect to K' must also be simultaneous with respect to K. In §2 he shows that, as a consequence, there is no *a priori* reason to maintain the classical identification $x' = x$. Since spatial distances depend on the adopted notion of simultaneity, the length of a moving rod as measured in the co-moving system K' is not necessarily equal to the rest length of that rod as measured in the rest frame K.

Once these two prejudices are abandoned, it becomes possible to consider linear transformations between coordinate systems in uniform parallel translation that differ from the classical ones, for which $t' = t$ and $x' = x - vt$. In §3 Einstein then seeks a new set of coordinate transformations that leave the velocity of light c invariant. The derivation is somewhat cumbersome (Martinez, 2009, 320–354), but Einstein nevertheless arrives at the so-called Lorentz transformations for motion along the common x–x' axis:

$$x' = \gamma(x - vt), \qquad t' = \gamma\left(t - \frac{v}{c^2}x\right), \qquad \gamma = \frac{1}{\sqrt{1 - \frac{v^2}{c^2}}}.$$

The transformation appears here for the first time in its modern form, which makes ~~its perfect symmetry in x and t explicit~~ explicit the reciprocal mixing of the spatial and temporal coordinates, especially when time is written as ct.[2]

Finally, in the concluding sections of the kinematical part, Einstein derives the physical consequences of the Lorentz transformations. Once coordinates are interpreted as being measured by means of rods and clocks, the theory yields specific predictions about the behavior of such systems that are, in principle, accessible to experiment. In §4 he derives both length contraction and time dilation. In §5 Einstein then derives the relativistic law of velocity addition, which resolves the tension between the principle of relativity and the light postulate. While length contraction had already been discussed in Lorentz's earlier work, time dilation appears here as a genuinely new physical effect. Although Joseph Larmor and Lorentz had already obtained related formal results (Brown, 2005, sec. 4.4 and 4.5), Einstein gave clock retardation its distinctive physical interpretation as a universal consequence of the new kinematics, applying to any suitable clock. Indeed, notwithstanding the well-known priority disputes surrounding the 'discovery' of the special theory of relativity, ~~there is broad agreement~~ the argument can be made that the introduction of time dilation as a ~~physical~~ universal effect constitutes Einstein's distinctive contribution (Rindler, 1970). It would therefore be appropriate—although this is not customary—to speak of 'Einstein dilation' as the natural counterpart to 'Lorentz contraction'.

1.1 Time Dilation and the Relativistic Clock Retardation

As one might expect, the young Einstein's derivation of time dilation relies purely on algebraic manipulation.[3] He considers a clock that is permanently at rest in the moving system K'. This means that its spatial coordinate in K' is constant and, in particular, that $x' = 0$ for all times. The clock, however, moves with respect to K

[2] Although Einstein originally used β to denote the Lorentz factor, in what follows I adopt the modern convention, reserving γ for the Lorentz factor (γ^{-1} for the inverse Lorentz factor) and β for the dimensionless velocity $\beta = v/c$.

[3] This is one of the reasons I prefer to avoid the adjective 'geometrical' in this context.

with constant velocity v along the x-axis. Einstein therefore makes implicit use of the Lorentz transformation relating the spatial coordinates of K and K'.

$$x' = \gamma(x - vt),$$

Since $x' = 0$, this condition immediately yields

$$0 = \gamma(x - vt)\,.$$

Formally, a product vanishes if and only if at least one of its factors vanishes. Since $\gamma \neq 0$, it follows that

$$x = vt\,,$$

which is simply the equation of uniform motion with velocity v in the system K: a clock at rest in K' moves uniformly with velocity v in K. The time indicated by this clock is denoted by t', which is related to t by the Lorentz time-transformation equation:

$$t' = \gamma\left(t - \frac{v}{c^2}x\right)\,.$$

Substituting $x = vt$, one obtains

$$\begin{aligned} t' &= \frac{1}{\sqrt{1-\frac{v^2}{c^2}}}\left(t - \frac{v^2}{c^2}\,t\right) \\ &= \frac{1}{\sqrt{1-\frac{v^2}{c^2}}}\left(1 - \frac{v^2}{c^2}\right)t \\ &= t\sqrt{1-\frac{v^2}{c^2}}. \end{aligned}$$

It follows that the coordinate time interval Δt~~measured~~, read off by the synchronized lattice of clocks at rest in K between two successive ticks of a clock moving with respect to K, is longer by a factor ~~equal to the Lorentz factor~~ γ than the ~~corresponding coordinate~~ proper time interval $\Delta t'$ measured by ~~the same clock at rest in the~~ that same clock in its co-moving system K'.

Einstein suggests, albeit rather implicitly, that for velocities small compared with c, one may expand the square root in a Taylor series about $v/c = 0$ (i.e., the Maclaurin series in the dimensionless parameter $(v/c)^2$):

$$\sqrt{1-\frac{v^2}{c^2}} = 1 - \frac{1}{2}\frac{v^2}{c^2} - \frac{1}{8}\frac{v^4}{c^4} - \cdots\,.$$

Keeping only the leading correction and neglecting terms of fourth and higher order in v/c, one obtains

$$1 - \sqrt{1-\frac{v^2}{c^2}} \approx \frac{1}{2}\frac{v^2}{c^2}.$$

Hence the retardation per unit time is approximately

$$\frac{\Delta t}{t} \approx \frac{1}{2}\left(\frac{v}{c}\right)^2.$$

The longer the clocks run, the larger the difference between their readings as judged in the frame K. The retardation is second order in v/c. For ordinary velocities ($v \ll c$), the quantity v^2/c^2 is extremely small, so the retardation accumulates only very slowly. However, by expressing the effect *per second*, Einstein emphasizes that the deviation is systematic and cumulative. A precise quantitative prediction, however small, can be tested if either the velocity is sufficiently high, or the observation time is sufficiently long, or both.

1.2 The Longitudinal and Transverse Doppler Effects

After deriving "the required propositions of the kinematics that correspond to our two principles", Einstein proceeds "to show their *application* in electrodynamics" (Einstein, 1905b, 907). The sharp separation between the kinematical part of the paper and the subsequent dynamical parts (involving electromagnetism and particle dynamics) is one of the most characteristic features of Einstein's approach to the electrodynamics of moving bodies. The kinematical framework is established independently of dynamics, and in particular independently of Maxwell's equations; only afterward is it brought to bear on dynamical theories. The second part of the paper accordingly 'applies' the new kinematics to well-established dynamical laws in order to test whether they remain unchanged under the Lorentz transformations. The procedure consists in formulating the dynamical equations with reference to the 'moving' system K', performing the Lorentz transformation, and verifying whether the transformed equations retain the same form in the 'rest' system K. If they do not, the equations must be modified, and such modifications may entail empirically testable consequences.

In §6 Einstein shows that Maxwell–Hertz's equations already satisfy this criterion. From his point of view, the fact that no additional terms need to be introduced is a fortunate circumstance, since the Lorentz transformations were not derived from Maxwell's equations themselves. In §7 Einstein applies the same procedure to the description of optical plane waves. The Lorentz invariance of the wave equation entails modified relations between frequency, direction of propagation, and relative motion. This yields a unified derivation of the aberration of light and the relativistic Doppler effect as consequences of the Lorentz transformations. Both aberration and the longitudinal Doppler effect were already known in classical optics; however, Einstein's treatment provides their exact relativistic form, valid to all orders in v/c, without recourse to ether-based dynamical hypotheses. Moreover, Einstein predicts the transverse Doppler effect, a genuinely new phenomenon absent from classical theory, which arises solely from time dilation and has no classical analogue.

Einstein's derivation proceeds as follows ~~(Einstein, 1905b, 910–911)~~(Einstein, 1905b, 910–11). Consider a moving, non-accelerated system K' in which a very distant light source is at rest. In this system, the emitted radiation can locally be represented as a plane wave, that is, by sinusoidal electric and magnetic field components whose phase

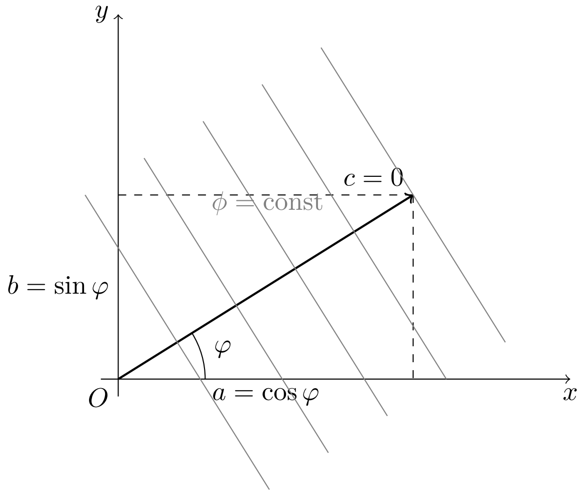

Figure 1: Plane wave propagating in the x–y plane

$$\phi' = \omega' \left[t' - \frac{a'x' + b'y' + c'z'}{c} \right]$$

defines a family of parallel phase planes $\phi' = \text{const}$ propagating rigidly through space. Here $\omega' = 2\pi\nu'$ is the angular frequency in the rest frame of the source, and (a', b', c') are the direction cosines of the wave normal. Without loss of generality, if the wave propagates in the x–y plane, one may set $a' = \cos\varphi$, $b' = \sin\varphi$, and $c' = 0$, where φ is the angle between the direction of propagation and the x-axis.

According to the principle of relativity, the same physical wave must also be describable in the rest system K with respect to which K' moves with uniform velocity. Einstein assumes that the wave preserves its plane-wave character in K, so that it can again be represented by a phase of the same functional form,

$$\phi = \omega \left[t - \frac{ax + by + cz}{c} \right].$$

Since the wavefronts themselves are physical entities, one and the same wavefront must correspond to the same value of the phase in all inertial frames (Einstein, 1905b, 911). This requirement is expressed by the condition

$$\phi' = \phi.$$

To implement this condition, the phase ϕ' must be rewritten in terms of the unprimed variables x, y, z, t. That is, the mathematical description of the wave given in K' must be expressed using the coordinates of K. This is achieved by substituting the Lorentz transformations for the time coordinate and for the longitudinal coordinate x, together with $y = y'$ and $z = z'$, into the expression for ϕ'. Upon regrouping terms, the phase again retains the same functional form, but with transformed coefficients:

$$a = \frac{a' + \frac{v}{c}}{1 + \frac{v}{c}a'}, \qquad b = \frac{b'}{\gamma\big(1 + \frac{v}{c}a'\big)}, \qquad c = \frac{c'}{\gamma\big(1 + \frac{v}{c}a'\big)}, \qquad \omega = \gamma\,\omega'\Big(1 + \frac{v}{c}a'\Big)\,.$$

As mentioned, a, b, c are the direction cosines of the wave normal, that is, the components of the direction of propagation (fig. 1). Their transformation therefore expresses the change in the apparent direction of propagation between K' and K. By definition, the angular frequency ω of a plane wave is given by the coefficient of the time coordinate in the phase when the spatial coordinates are held fixed. Accordingly, the coefficient multiplying t in the transformed phase determines the observed angular frequency ω in the system K. The invariance of the phase thus entails two distinct but closely related effects. The transformation of the temporal coefficient yields a change in frequency, that is, the relativistic Doppler effect, whereas the transformation of the spatial coefficients encodes a change in the apparent direction of propagation of the wave, that is, the aberration of light. The Doppler effect and aberration therefore arise simultaneously from the same Lorentz transformation of the phase, as two aspects of a single kinematical procedure.

We are here interested in the frequency shift (Einstein, 1905b, 911). Transforming the phase of the wave from K' to K and passing from angular to ordinary frequency, $\nu = \omega/(2\pi)$, one obtains the relativistic Doppler formula

$$\nu = \nu'\,\frac{1 + \frac{v}{c}\cos\varphi}{\sqrt{1 - \frac{v^2}{c^2}}}.$$

For longitudinal propagation along the x–axis, one has $\varphi = 0$ and hence $\cos\varphi = 1$. The general expression then reduces to the familiar relativistic formula for the longitudinal Doppler effect,

$$\nu = \nu'\,\frac{1 + \frac{v}{c}}{\sqrt{1 - \frac{v^2}{c^2}}} = \nu'\,\sqrt{\frac{1 + \frac{v}{c}}{1 - \frac{v}{c}}}.$$

As this shows, when one starts from a source at rest in the moving system K' and evaluates the frequency in the system K, the longitudinal Doppler effect combines a classical Doppler factor, $1 + \frac{v}{c}$, associated with relative motion along the line of sight, with an additional relativistic correction $1/\sqrt{1 - \frac{v^2}{c^2}}$. In classical wave theory, if the source is approaching the observer, the wave crests arrive more frequently, and the observed frequency is therefore higher than that of an identical source at rest. Relativity predicts a further frequency shift beyond this classical effect.

The case $\cos\varphi = 0$ corresponds to motion perpendicular to the direction of propagation of the light, that is, to transverse motion (along the y–axis in fig. 1). In this case the formula reduces to

$$\nu = \frac{\nu'}{\sqrt{1 - \frac{v^2}{c^2}}},$$

showing that even when there is no component of the relative motion along the line of sight, the observed frequency nevertheless differs from the emitted one. A frequency shift therefore appears even though there is no classical Doppler effect in the Galilean theory of waves. This purely relativistic transverse Doppler effect has no analogue in pre-relativistic wave theory.

2 Testing the New Kinematics: Stark's Second-Order Doppler Effect

In the 1905 paper, time dilation and the transverse Doppler effect appear, respectively, among the direct and indirect consequences of the Lorentz transformations. Einstein does not explicitly relate the two effects to one another, nor does he discuss the possibility of an empirical test of the transverse Doppler effect. He likely did not think that wave sources were available that could attain velocities large enough for such a small effect to be experimentally accessible. Rather, in §8 of his relativity paper, Einstein derives the transformation law for the frequency of light and argues that the energy of light transforms in the same way as its frequency, i.e. $\frac{E'}{E} = \frac{\nu'}{\nu}$. On this basis, he derives the radiation pressure exerted by light on a perfectly reflecting mirror in uniform motion. This result is subsequently exploited in the September paper on the inertia of energy (Einstein, 1905a). In §9 Einstein then turns to the Maxwell equations with sources. The only explicit discussion of the empirical testability of the new theory occurs in §10 of the paper. In this section Einstein derives the equation of motion of the slowly moving 'electron', an expression that he uses to denote massive charged particles in general. Insofar as electrons (in the sense of elementary particles) function as fast-moving charged test particles, the predicted effects are, in principle, accessible to observation.

Indeed, Einstein's 1905 relativity paper was cited for the first time on November of 1905 by the Göttingen experimentalist Walther Kaufmann (1905, 954), in a short note presented at a plenary session of the Prussian Academy of Sciences. The note reported on Kaufmann's new experiments with Becquerel rays, or β-rays, emitted by radioactive substances. As is well known, Kaufmann's results appeared unfavorable to relativity, a conclusion he articulated more fully in a longer essay published at the beginning of 1906 (Kaufmann, 1906). The data aligned more closely with Abraham's theory, according to which the electron was assumed to be non-deformable. The issue thus seemed to be settled against the Lorentz–Einstein electron in favor of Abraham's spherical model. However, in March Planck (1906a) urged caution. The results of Kaufmann's experiments were not sufficiently precise to warrant a definitive refutation.

According to Planck (1906b), relativistic dynamics had the significant advantage of being derived without reliance on any specific model of the electron and was therefore worth exploring. At this juncture, the young Einstein himself clearly regarded experiments on fast-moving electrons as the only means of testing the theory. In August 1906 he submitted a paper (Einstein, 1906) proposing a new experimental method to determine the ratio of transverse to longitudinal electron mass at high velocities using electrostatic deflection alone. The paper was published in issue 21 of the *Annalen der Physik*. In the same issue, a contribution by Johannes Stark (1906), submitted in September 1906, also appeared. It is therefore plausible that Einstein

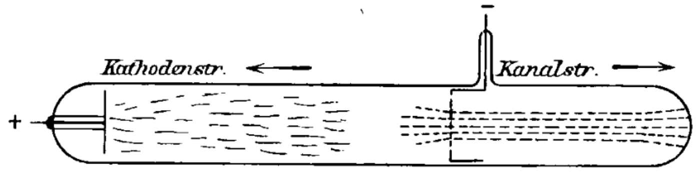

Figure 2: Discharge tube producing cathode rays and canal rays (Stark, 1906, 405)

read Stark's paper upon receiving his copy of the *Annalen* issue. He may then have realized that wave sources were indeed available to test the transverse part of the relativistic Doppler effect.

2.1 Stark's Canal-Ray Experiments and Velocity-Dependent Spectral Effects

Stark could rely on decades of spectroscopic work showing that excited gases and vapors do not produce continuous spectra, but rather sharp spectral lines appearing at well-defined positions in a spectrograph (Hentschel, 2002). The Zeeman splitting of such lines in a magnetic field was found to be consistent with a charge-to-mass ratio equal to that of the electron, and the sense of the splitting indicated a negative charge. Within the framework of Lorentz's electron theory, it was therefore concluded that the 'centers' (*Zentren*) of emission responsible for line spectra are negatively charged oscillating electrons (Stark, 1906, 401). Stark's central aim was to identify the physical systems that serve as the 'carriers' (*Träger*) of different spectra, that is, the material systems bearing the electronic oscillatory processes responsible for line emission (Stark, 1906, 402). He conceived the carrier as a positive atomic ion containing bound negative electrons executing elastic oscillations about equilibrium positions. Radiation was attributed to these accelerated charges, and spectral lines were identified with their characteristic oscillation frequencies; the sharpness and reproducibility of the lines were taken as empirical evidence for stable intrinsic atomic periodicities. The distinction between 'centers' and 'carriers' is crucial to Stark's theory, since it allows the physical state of the carrier to influence the behavior of the oscillating centers.

Stark's experimental investigations begin with a gas-filled discharge tube, typically containing hydrogen (fig. 2). When an electrical discharge is applied, the discharge current is largely carried by fast negative electrons, which are accelerated by the electric field toward the anode. As these electrons traverse the gas, they undergo collisions with neutral atoms or molecules. In sufficiently energetic collisions, electrons are knocked out of atoms, producing positive ions, that is, "atoms that have lost one or several negative electrons through ionization" (Stark, 1906, 402). These ions are accelerated toward the cathode through a potential drop V concentrated in the cathode fall, which sets the velocity scale of the moving ions. If the cathode is perforated, a fraction of the ions passes through the holes and continues beyond the cathode with approximately rectilinear motion, forming a beam propagating away from it. These beams of positive ions are known as canal rays (*Kanalstrahlen*), named after the 'canals' in the cathode through which they pass ~~(Stark, 1906, 403)~~(Stark, 1906, 403).

When accelerated toward the cathode, the positive ions emit characteristic spectral radiation in all directions as a result of excitation processes in the discharge. Stark analyzed this radiation using a prism spectrograph. The emitted light entered the instrument through a narrow slit, which selected a well-defined direction of propagation (longitudinal or transverse with respect to the ion motion). After passing through the slit, the light is rendered approximately parallel and passed through a glass prism, which ~~separates~~ separated different wavelengths λ by refraction; the dispersed radiation is finally recorded on a photographic plate. In the case of hydrogen, Stark took the Balmer series as empirically given, that is, as a set of well-defined spectral lines Hα, Hγ, ... (Stark, 1906, 417). Whether the emitting system consisted of neutral hydrogen in a discharge tube or of hydrogen ions in canal rays, the same characteristic lines reappeared. When the emitting ions moved along the line of sight of the spectrograph (toward b or c in fig. 3), Stark observed that the entire pattern of spectral lines was displaced toward longer wavelengths (redshift) for receding motion (toward c), and toward shorter wavelengths (blueshift) for approaching motion (toward b). The magnitude of this Doppler displacement increased with the ratio v/c:

$$\frac{\Delta\lambda}{\lambda} \sim \pm\frac{v}{c}.$$

Stark concluded that sharp line spectra (*Linienspektrum*) are emitted by individual atomic ions in translational motion. By contrast, the *band spectrum* (*Bandenspektrum*)—consisting of groups of closely spaced lines forming characteristic bands—exhibited no measurable Doppler displacement. Stark therefore attributed band spectra to neutral atoms or molecular aggregates formed in recombination processes, whose translational velocities are much smaller than those of the canal-ray ions (Stark, 1906, 426).

In the first two parts of his paper, Stark established the ordinary first-order Doppler effect proportional to v/c and concluded that "[t]he band spectrum and the line spectrum of hydrogen do not have the same carrier" (Stark, 1906, 414). In part III, he raised a further question: whether an additional displacement of spectral lines might exist that depends only on the magnitude of the ion's velocity rather than on its direction. Any such effect would necessarily be proportional to v^2/c^2, since linear terms change sign when the direction of motion is reversed, whereas quadratic terms do not (Stark, 1906, 409–10):

$$\frac{\Delta\lambda}{\lambda} \sim \frac{1}{2}\frac{v^2}{c^2}.$$

To test for such a direction-independent effect, Stark arranged observations perpendicular to the direction of ion motion (the spectrograph a in fig. 3). In this transverse configuration, the classical longitudinal Doppler effect vanishes by symmetry, since there is no velocity component along the line of sight.

Stark interpreted any residual shift as potentially dynamical in origin, arising from a velocity-dependent "modification of the electromagnetic forces between the electrons, which is measured by even powers of the ratio v/c" (Stark, 1906, 449). In Stark's model, the ion as a 'carrier' is not a rigid body but a physically deformable system, whose internal state can change with its motion and environment (Stark, 1906, 449). As the translational velocity of the ion increases, the amplitudes of the internal 'centers' of emission (the electrons) were assumed to increase, since the strong electric fields required to accelerate the ions were thought to excite the bound electrons into

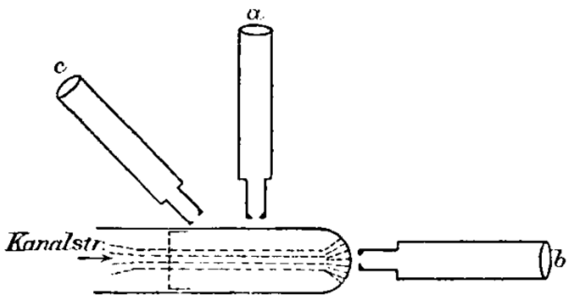

Figure 3: ~~Set up~~ Setup for observing the Doppler shift of canal-ray spectral lines, with spectrographs aligned parallel and perpendicular to the direction of propagation (Stark, 1906, 410)

oscillations of larger amplitude. Moreover, Stark speculated that radiation pressure acting on the oscillating electrons, the finite propagation speed of electromagnetic interactions, or deformation of the oscillatory system at high velocities might contribute to such effects. On this view, the combined action of increased electron amplitudes and radiation pressure could produce a physical deformation of the carrier, which Stark investigated as a possible mechanistic cause of wavelength shifts.

Stark reported tentative empirical indications consistent with this picture, noting a ~~red shift~~ redshift of the midpoints of certain spectral lines with increasing ion velocity—for example, a displacement of about 0.42 Å for the Hγ line at a cathode fall of 7500 Volts (Stark, 1906, 450). He emphasized, however, that while the attainable ion velocities were already sufficient to produce an observable longitudinal Doppler effect, which is first order in v/c, they were still insufficient, in practice, for the transverse effect proportional to v^2/c^2 to be unambiguously isolated. In addition, he stressed the extreme experimental difficulty of ensuring exact transverseness. The slightest radial component would swamp the transverse effect. In particular, he calculated that a misalignment of only $1°59'$ would suffice to generate a spurious ~~red shift~~ redshift of comparable magnitude through the ordinary longitudinal Doppler effect. For these reasons, Stark explicitly characterized these observations as orienting indications rather than as conclusive experimental confirmation (Stark, 1906, 450–51).

2.2 Einstein and Stark's Second-Order Effect as a Test of Time Dilation

At that time, Einstein's new dynamics of free electrons was under attack. The Einstein–Planck approach to electron dynamics was criticized by Ehrenfest (1907): since ~~no(Einstein, 1907b, 197)n-spherical~~ non-spherical electrons would, in principle, behave differently from spherical ones, the absence of a definite electron model was taken to be a defect rather than a virtue. Einstein (1907a), however, was not particularly impressed by this objection. He famously emphasized that relativity was not a complete theory, but merely imposed kinematical constraints on admissible theories. Einstein did not develop a theory of the electron, but modified the equations of motion of a charged particle in general so as to make them Lorentz invariant. After coming across Stark's (1906) paper toward the end of 1906, Einstein may have quickly realized that this work opened the possibility of a more direct test of the new kinematics itself, without recourse to electron dynamics. He addressed this issue in a paper submitted on March of 1907.

Stark's experiments showed that wave sources moving at very high velocities were in fact available: the positive ions forming canal rays emit line spectra that exhibit a longitudinal ~~first order~~ first-order Doppler effect, that is, an effect proportional to v/c due to their motion. As we have seen, Stark also attempted to detect an additional ~~second order~~ second-order effect, proportional to v^2/c^2; however, the experimental setup, not specifically designed for this purpose, was insufficient to yield a reliable result. Nevertheless, it was shown that, in principle, it was possible to decide whether the spectral lines of ~~fast moving~~ fast-moving ions exhibit a redshift when they are observed spectroscopically at right angles to the direction of motion. Einstein's aim was to show that the principle of relativity, together with the principle of the constancy of the velocity of light, suffices to account for the effect in question (Einstein, 1907b, 197). It is worth noting, however, that Einstein did not present Stark's work as a test of the relativistic wave equation as a whole, but rather as a test specifically of time dilation, since in the transverse case the effect depends solely on the time transformation.

Referring to his earlier 1905 paper, Einstein reminds his readers that, in relativity theory, a uniformly moving clock runs more slowly, when judged from a 'stationary' system~~, than when judged~~ : successive ticks of a clock at rest in K' are assigned a longer coordinate-time interval by the synchronized clocks at rest in a 'stationary' system K than by an observer co-moving with the clock. Einstein appears implicitly to assume that any periodic system passing through identical phases can be regarded as a clock. If ν_0 denotes the frequency defined as the number of periods N divided by the time interval $\Delta t'$ measured in the co-moving system K', and ν denotes the frequency defined as the same number N divided by the coordinate time interval Δt measured in a system K relative to which the atom is moving, then, since $\Delta t = \gamma\,\Delta t'$, special relativity predicts:

$$\frac{\nu}{\nu_0} = \sqrt{1 - \frac{v^2}{c^2}}, \tag{1}$$

Testing this relation requires clocks that can be accelerated to very high velocities without compromising their accuracy. Direct experimental verification using mechanical clocks is therefore likely to remain unattainable, since the velocities that can be imparted to such devices are negligible in comparison with the speed of light. Nature, however, provides systems that effectively function as clocks and that can be accelerated, by means of an electric field, to extremely high speeds: "The atom ion of the canal rays, which emits and absorbs radiation of definite frequencies, is therefore to be regarded as a rapidly moving clock, and the relation just given is thus applicable to it" (Einstein, 1907b, 198).

One might object that the frequency ν_0 of the fast-moving ion cannot be measured by a co-moving spectrograph. However, according to Einstein, there was sufficient evidence to support the claim that, whether the emitting system consisted of neutral hydrogen in a discharge tube or of hydrogen ions produced in canal rays, the same characteristic spectral lines reappeared. This was taken as strong evidence that the Balmer series expresses a stable property of the hydrogen system, rather than something contingent on the manner in which it was produced. Einstein could then point out that the corresponding frequencies are determined solely by the nature of the ion—that is, ions of the same species yield the same frequency ν_0 when measured with a spectrograph at relative rest:

> However, one has to take into consideration that the frequency ν_0 (for the co-moving observer) is unknown, so that the above relation is not accessible to direct experimental verification. But, it may be assumed that ν_0 is also equal to the frequency emitted or absorbed by the same ion while at rest, and this for the following reason. From the fact that one and the same line spectrum is formed under very different conditions, we conclude that the frequency ν_0 does not depend on interactions between moving ions and the stationary gas, but is a characteristic of the ion only; from this one directly concludes with the help of the principle of relativity that ν_0 must equal the frequency of radiation emitted or absorbed by an ion at rest. (Einstein, 1907b, 198)

Einstein then assumes that ions of a given species can be treated as identical systems: the recurrence of the same sharp spectral lines under widely varying experimental conditions supports the assumption that their characteristic frequencies ν_0 are intrinsic properties of the ions themselves and do not depend on interactions with the surrounding medium. That is, all ions of the same species exhibit the same spectral lines when at relative rest. Since canal-ray ions move at very high velocities, any possible shift ν of the spectral lines with respect to ν_0 would thus become, at least in principle, empirically accessible.

The ~~first–order~~ first-order Doppler effect arises from the relative motion between source and observer and depends linearly on v/c. In classical theory it does not presuppose any particular microphysical account of the emission process, nor does it require modifications of the internal dynamics of the emitter. In Stark's work, however, the ~~second–order~~ second-order Doppler effect—that is, any frequency shift proportional to v^2/c^2 observed in a transverse configuration—could not be explained by simple relative motion along the line of sight. Stark therefore treated the effect as a genuinely dynamical modification of the 'carrier': he assumed that uniform motion produces a real change in the emission process of fast ions—a kind of reduction in the number N of oscillations completed ~~during a *same time*~~ by the moving ion during the same universal time interval (frequency contraction). Einstein explicitly rejected this dynamical explanation. Instead, he invoked the principle of relativity to support the assumption that uniform motion does *not* alter the internal oscillatory processes of the ion~~; the number~~. The same emission process involves the same count N of ~~oscillations associated with a given emission process is the same in all non-accelerated systems, but spread over *different* time intervals time dilation)~~completed oscillations; the relativistic effect arises because this same count is referred to different time intervals in the two inertial descriptions.

This assumption is the key to Einstein's reasoning. The ~~number N of oscillations is~~ intrinsic frequency $\nu_0 = N/\Delta t'$ is taken to be fixed, and longitudinal Doppler components are excluded by definition. As a consequence, the remaining transverse Doppler effect, that is, the reduced observed frequency $\nu = N/\Delta t = \nu_0/\gamma$, can *only* be attributed to the transformation properties of the time coordinate between inertial reference systems in uniform relative motion, $\Delta t = \gamma \Delta t'$, as prescribed by Einstein's new kinematics. The transverse Doppler effect thus serves as a test of the ~~time–dilation~~ time-dilation relation. Expanding eq. (1) in a series, Einstein (1907b, 197) obtains, to second order in v/c,

$$\frac{\nu - \nu'}{\nu'} \approx -\frac{1}{2}\left(\frac{v}{c}\right)^2 , \tag{2}$$

which quantifies this deficit. Equation (2) therefore provides a direct theoretical expression for the sought ~~second order~~ second-order effect in the radiation emitted by fast ~~canal ray~~ canal-ray ions.

Equation (2) can be tested experimentally by means of a spectrograph oriented such that the motion of the emitting ions is perpendicular to the line of sight; in this configuration, all ~~first order~~ first-order Doppler contributions vanish. A spectrograph registers the wavelength at which a given spectral line appears after dispersion. The observed position of a spectral line therefore determines its wavelength λ, and, via the relation $\lambda = c/\nu$, its frequency ν. By comparing the wavelength λ of the line emitted by the moving ions with the reference wavelength λ_0 measured for ions at rest, the spectrograph allows one to determine the fractional shift $(\lambda - \lambda_0)/\lambda_0$, which is equivalent (up to a change of sign) to $(\nu - \nu_0)/\nu_0$. Comparing the relativistic prediction with Stark's experimental results, Einstein noted that the numerical values reported by Stark exceeded the expected effect by more than an order of magnitude. However, he expressed skepticism regarding the reliability of the existing measurements (Einstein, 1907b, 198). Nevertheless, in correspondence with Stark in April 1907, Einstein expressed the hope that it might eventually become possible to detect such ~~second order~~ second-order effects experimentally with greater accuracy (Einstein to Stark, Apr. 13, 1907; Einstein, 1987, Vol. 5, Doc. 45).

2.3 The Transverse Doppler Effect in Einstein's 1907/1908 Jahrbuch Paper

Stark was the editor of the *Jahrbuch der Radioaktivität und Elektronik*. In September, he invited Einstein to contribute a comprehensive review article on the theory of relativity, and Einstein agreed to "deliver the desired report" (Einstein to Stark, Sep. 25, 1907; Einstein, 1987, Vol. 5, Doc. 58). Stark assisted him by supplying relevant literature that the young Einstein could not easily access (Einstein to Stark, Oct. 7, 1907; Einstein, 1987, Vol. 5, Doc. 61). The paper was ultimately submitted on December 4, 1907 and appeared in the *Jahrbuch* at the end of ~~.~~ January 1908. Beyond synthesizing his earlier relativity work, ~~the~~ Einstein (1908) incorporated several recent developments, including Laue's (1907) derivation of the drag coefficient, Planck's (1906a) reformulation of relativistic dynamics, and related advances in the treatment of thermodynamics within the relativistic framework (Planck, 1907). As expected, Einstein also used the 1908 review to discuss Stark's (1906) observations as a possible means of testing the new kinematics.

A distinctive feature of Einstein's new presentation is that the transverse Doppler effect is no longer treated primarily as a consequence of the Lorentz-invariant wave equation, but is instead relocated within the kinematical framework of the theory. Stark's work on the second-order transverse Doppler effect is mentioned explicitly in the kinematical part I of Einstein's paper, where time dilation is discussed as one of the consequences of the new kinematics (Einstein, 1908, 422). In the electrodynamic part II, by contrast, Einstein derives the Lorentz-invariant wave equation and hence the relativistic Doppler effect without explicitly distinguishing between longitudinal and transverse cases. Although Stark is not mentioned in this context, Einstein does point out that the relativistic Doppler effect can be tested by measuring the frequency shift of light emitted or absorbed by canal rays, which depends both on the velocity of the ions and on the direction in which the radiation is observed (Einstein, 1907a, 425). At the time, canal rays were essentially the only laboratory sources capable of reaching velocities sufficiently large for such effects to be spectroscopically accessible.

In this way, the discussion of the transverse Doppler effect is conceptually shifted from the domain of optics and electrodynamics to that of kinematics ~~(Einstein, 1908, 421-22)~~(Einstein, 1908, 421–22). The presentation of the time-dilation effect here differs significantly from that of the 1905 paper: it places far less emphasis on algebraic derivation and is instead oriented toward clarifying the physical interpretation and the prospective experimental testability of the effect. Einstein ~~considers again~~ again considers a clock U' permanently at rest at $x' = 0$, at the origin of the moving system K', which moves uniformly with velocity v relative to K along the x-axis. The clock is modeled as a physical periodic system with an intrinsic rest frequency ν_0. The intrinsic frequency ν_0 characterizes the clock as a physical system and is assumed to be independent of its state of uniform motion. Accordingly, the successive ticks of the clock ~~are invariantly~~ can be labeled by the integers $n = 1, 2, 3, \ldots$, corresponding to the completion of successive periods of the same physical process. ~~In this sense, the number N of ticks associated with a given physical process used as a clock is assumed from the outset to be invariant.~~

Since the clock U' is at rest in K', that is, since $\Delta x' = 0$, its successive ticks occur at equal intervals of the time coordinate t':

$$t'_n = \frac{n}{\nu_0},$$

where the relation between the discrete tick number n and the continuous time coordinate t' is fixed by the convention that assigns the rate ν_0 to the clock. Accordingly,[4] the time instants at which successive periods are completed in the system K' are

$$t'_1 = \frac{1}{\nu_0}, \quad t'_2 = \frac{2}{\nu_0}, \quad t'_3 = \frac{3}{\nu_0}, \quad \cdots \quad t'_n = \frac{n}{\nu_0}.$$

Einstein then asks how fast this same clock runs when viewed from the system K. To answer this question, he determines the coordinate times in K at which the successive periods of the moving clock are completed. These times are obtained by applying the Lorentz transformation to the time coordinate of K':

$$t = \gamma \left(t' - \frac{v}{c^2} x' \right).$$

Since the clock U' remains permanently at rest at the origin of K', one always has $x' = 0$. The coordinate time in K corresponding to the completion of the nth period

[4] In the following I also rely on ~~(Einstein, 1910, 131-32)~~ Einstein, 1910, 131–32 to clarify Einstein's otherwise somewhat elliptical reasoning.

is therefore

$$t_n = \gamma t'_n = \frac{\gamma}{\nu_0}\, n.$$

Accordingly, the successive periods of the moving clock U' are completed, in the system K, at the time instants

$$t_1 = \gamma\,\frac{1}{\nu_0}, \quad t_2 = \gamma\,\frac{2}{\nu_0}, \quad t_3 = \gamma\,\frac{3}{\nu_0}, \quad \cdots \quad t_n = \gamma\,\frac{n}{\nu_0}.$$

Comparing the coordinate-time differences $t_n - t_1$ and $t'_n - t'_1$ assigned to the same ~~1rs~~ first and nth tick of U' in the two systems K and K', one finds that

$$\Delta t = \gamma\,\Delta t'.$$

The frequency of the clock U' at rest in K' is ν_0. When the same clock is described from the system K, the same number N of periods is distributed over a longer elapsed coordinate-time interval $\Delta t = \gamma\,\Delta t'$. From the standpoint of K, U' ~~appears therefore to have~~ therefore appears to have a reduced frequency ν:

$$\nu = \frac{N}{\Delta t} = \frac{N}{\gamma\,\Delta t'} = \nu_0\sqrt{1 - \frac{v^2}{c^2}}. \tag{3}$$

In this precise sense, one can say that the moving clock runs 'more slowly' than an identical clock at rest in K: the ticks of the moving clock are assigned a longer coordinate-time interval by the synchronized clocks at rest in K than the ticks of an identical clock at rest in K.

~~This statement~~The claim that the moving clock runs 'more slowly', however, must be treated with some caution. As one can see, Einstein's argument rests on the fundamental assumption that clocks are *physically identical* across ~~non-accelerated~~ non-accelerated reference frames, that is, that they possess the same intrinsic frequency ~~ν_0~~ $\nu_0 = N/\Delta t'$ independently of their uniform motion. Under this assumption, ~~the number N of periods is taken to be invariant, and~~ time dilation is attributed ~~entirely to the expansion of the elapsed~~ to the dilation of the coordinate-time interval Δt assigned in K to the same number N of completed periods of the moving clock. One of the most significant novelties of Einstein's 1908 paper is that he made this assumption explicit for the first time: "the rate of a clock does not undergo any permanent change if such objects are set in motion and then brought to rest again" (Einstein, 1908, 420; fn. 1). If the clock U' were carefully brought to rest and placed next to any of the clocks U at rest in K, the two would run at the same rate ν_0. This requirement may therefore be taken as a criterion for a 'good' clock, namely one whose rate is not permanently affected by its past history.

To test the relation eq. (3) empirically, one requires the existence of a physical system that actually behaves in this way, that is, a real clock capable of reproducing the same time units in different ~~non-accelerated~~ non-accelerated reference frames. *If* such a system exists, *then* the consequences of the new kinematics for time become empirically accessible. As we have seen, nature offers objects that possess the character of ideal clocks, namely ions emitting spectral lines:

> The formula $\nu = \nu_0 \sqrt{1 - \frac{v^2}{c^2}}$ admits of a very interesting application. Mr. J. Stark showed in the previous year that the ions forming canal rays emit line spectra, by observing a displacement of spectral lines that is to be interpreted as a Doppler effect. Since the oscillation process that corresponds to a spectral line is to be considered an intra-atomic process *whose frequency is determined by the ion alone*, we may consider such an ion as a clock of a certain frequency ν_0, which can be determined, for example, by investigating the light emitted by *identically constituted* ions which are at rest relative to the observer. The above consideration shows, then, that the effect of motion on the light frequency that is to be ascertained by the observer is not completely given by the [longitudinal] Doppler effect. The motion also reduces the (*apparent*) proper frequency of the emitting ions in accordance with the relation given above. (Einstein, 1908, 422; my emphasis)

This passage makes clear that the premise of the argument is that ions of the same species are taken to be physically identical when compared at relative rest; they are all 'identically constituted'. By 1908, spectroscopy had established with great precision that atoms and ions of the same species emit sharp, reproducible spectral lines whose frequencies are invariant across laboratories and experimental conditions, provided the emitters are compared at rest. This empirical stability strongly suggested that atomic emission frequencies ν_0 are intrinsic properties of the emitters, not contingent on macroscopic conditions or preparation histories. Despite the lack of a theoretical explanation, for Einstein this made the atom, the "producer of spectral lines" (Einstein, 1908, 459), uniquely well suited to function as a natural clock.

Since ions are assumed to be identical in all frames in uniform relative motion, the frequency shift of fast-moving ions measured by a spectrograph at rest cannot be attributed to a change in the ions themselves. In the longitudinal case, the observed shift may still be explained by motion along the line of sight. By contrast, for a spectrograph oriented perpendicular to the direction of motion, no such explanation is available. In this case, the observed redshift must be understood as a consequence of time dilation alone. For this reason, Einstein described the redshift as *apparent*.[5] The reduction in frequency is frame dependent and ~~disappears entirely for~~ is not recorded by a spectrograph co-moving with the ion. The adjective *apparent* was likely chosen to contrast the relativistic account with Stark's dynamical interpretation.

As we have seen in section 2.1, in Stark's view, the motion of the ion produces a *real* frequency contraction, that is, a modification of the emitting system's intrinsic oscillation frequency ν_0, such that ~~the ion completes a smaller number N of oscillations in the *same* time interval $\Delta t'$~~ fewer oscillations are completed during a given time interval in the moving frame. In Einstein's kinematical interpretation, by contrast, the intrinsic periodic process remains unchanged, since it is fixed by the same physical emission mechanism. The ~~number N of oscillations associated with a given emission process is invariant. The~~ observed redshift is therefore not due to a change in the emitter's frequency ν_0, but to time dilation: relative to a frame K in which the ion is moving~~the same number of oscillations N is~~, the same completed oscillations are distributed over a *longer* time interval $\Delta t = \gamma \Delta t'$ than in the rest frame K'. The frequency shift is thus explained by the relativistic transformation of the time coordinate, not by a dynamical modification of the source. However, Einstein clearly

[5] If a transversely moving source produces a redshift $\nu = \gamma^{-1}\nu_0$, then a transversely moving spectrograph relative to a source at rest produces the inverse effect, a blueshift $\nu = \gamma\nu_0$. The two situations are completely symmetrical in principle, but only the redshift from a transversely moving source is usually referred to as the transverse Doppler effect. While fast-moving atoms are readily available in canal rays, fast-moving spectrographs are not experimentally feasible. Nevertheless, this symmetry shows that the transverse Doppler effect cannot be attributed to a real, direction-dependent dynamical deformation of the atom.

regarded this 'apparent' effect as an unavoidable consequence of the theory and as one that could, in principle, be empirically detected.

Since the effect is of second order in v/c, it requires clocks moving at sufficiently high velocities, a condition satisfied by ions in canal rays. If two identical spectrographs are at rest with respect to the source but operate for different exposure times, the longer exposure merely collects a larger number N of wave crests: both the elapsed time and the number of accumulated oscillations increase proportionally. As a result, the inferred frequency ν_0 remains unchanged, and the spectral lines appear at the same position; only the intensity varies. In the relativistic case, by contrast, the crucial point is that the same physical emission process is described from two different non-accelerated frames. The same number of oscillations N associated with a given emission process is thus referred to different elapsed times, $\Delta t = \gamma \Delta t'$. The difference in the temporal interval over which the same oscillations N are distributed necessarily entails a change in the observed frequency ν. Through the relation $\lambda = c/\nu$, this difference in temporal rate is translated into a difference in wavelength, which the spectrograph at rest in K registers directly as a redshift of the atom's spectral line.

3 Ritz and the Transverse Doppler Effect as an *Experimentum Crucis*

Replying to a letter from Sommerfeld that is no longer extant, Einstein expresses strong interest in having a ~~canal ray~~ canal-ray experiment carried out and welcomes Sommerfeld's willingness to involve Peter Paul Koch (at that time an assistant of Wilhelm Röntgen in Munich) in testing the transverse Doppler effect. Einstein nevertheless betrays some caution as to whether Koch would actually carry out the experiments (Einstein to Sommerfeld, Jan. 14, 1908; Einstein, 1987, Vol. 5, Doc. 73). He notes that Stark had already informed him some months earlier of his intention to undertake the same investigation, although Stark had since written to Einstein repeatedly without again mentioning the experiment (Einstein to Sommerfeld, Jan. 14, 1908; Einstein, 1987, Vol. 5, Doc. 73). The letter to Sommerfeld also shows that this was something of a side issue. The main point of debate concerned the empirical test of the variability of the electron mass investigated by Kaufmann. Indeed, Einstein's 1908 paper devotes the entire §10 to a discussion of Kaufmann's experiments and to a comparison between the different predictions of the Abraham electron, the Einstein–Lorentz electron, and the Bucherer electron (Walter, 2018).

These electron models were all constructed within a theoretical context shaped by the assumed validity of Maxwellian electrodynamics and by constraints imposed by optical and electrodynamic phenomena, even though they differed markedly in their interpretations of the status and significance of the Lorentz transformations. A more radical challenge to relativistic kinematics was posed by Ritz, who in the winter of 1907–1908 moved from Göttingen to Tübingen, where he worked with the experimentalist Friedrich Paschen. Regarded as a rising star of European physics for his work in spectral theory (Hentschel, 2012), Ritz became increasingly skeptical of Lorentz–Maxwell electrodynamics after a stay in Leiden with his close friend Ehrenfest (Martinez, 2004). His major polemical paper appeared in the *Annales de chimie et de physique* in February 1908, only a few weeks after the publication of Einstein's 1908 review article. Like Einstein, Ritz (1908a, 147–48) was skeptical of the reliability of Kaufmann's experiments. However, for him, the difficulties involved in constructing a suitable model of the electron and in accounting for the velocity dependence of its mass were not isolated technical problems but rather symptoms of a deeper flaw in Maxwell–Lorentz electrodynamics itself.

Lorentz–Maxwell electrodynamics assumes the existence of an absolute system of ether coordinates, taken to be independent of the motions of matter, with respect to which light moves with velocity c. In order to reconcile this framework with the failure of the Michelson–Morley and other ether-drift experiments, it became necessary to eliminate the absolute system by means of increasingly implausible auxiliary hypotheses (Ritz, 1908a, 148). As Ritz complained, this strategy, pursued consistently by Lorentz and Einstein, required modifying the principles of kinematics themselves: abandoning absolute time and ~~rigid bodies, and~~ replacing 'rigid' with 'deformable' bodies, while also treating the parallelogram rule for the composition of velocities as a mere first approximation, valid only at low speeds. However, "[i]t would be regrettable, for the economy of our thinking, if we were forced to admit such complications" (Ritz, 1908a, 148). By abandoning Maxwellian electrodynamics in favor of a description based solely on retarded potentials, Ritz was led naturally to a ballistic theory of light, in which radiation propagates with the velocity of its source, $c \pm v$.

Ritz (1908c, 203) was well aware that the approaches of Lorentz and Einstein were not identical. Lorentz was committed to the reality (*réalité*) of length contraction and time dilation, whereas for Einstein lengths and times only appear (*semblera*) to be contracted and dilated, while true lengths and times remain invariable (Ritz, 1908c, 203). Nevertheless, this difference appeared to him to be marginal, since both approaches led to the same observable consequences ~~Ritz (1908b, 213)~~(Ritz, 1908b, 213). After returning to Göttingen in the spring of 1908, Ritz, in correspondence with Paschen and as a leading expert in spectroscopy, naturally singled out the transverse Doppler effect in canal rays as an experimentally decisive test case capable of discriminating between classical kinematics and the new kinematics~~.~~:

> When I was in Tübingen, you had a hydrogen tube in operation for canal-ray investigations. I would like to present to you a problem which is of the greatest importance for the question of the principle of relativity, and thus for the whole of electrodynamics. According to the Lorentz–Einstein theory of relativity, the wavelength emitted by a moving atom must change not only in the direction of motion in accordance with the Doppler principle; when observed perpendicular to the direction of the velocity, there must also occur a red shift of magnitude $\frac{1}{2}\left(\frac{v}{c}\right)^2\lambda$ (c = speed of light). For H_γ and $v = 1000$ km/sec, this would amount to about 0.02 Å.
>
> In observations of canal rays, one would therefore observe an apparent shift of the line by less than 0.02 Å, since the stationary and moving intensities could not, of course, be separated. With fine lines, however, and if one had the normal and the shifted spectrum on the same plate, this displacement could be detected even without a large grating. Would it not be possible to arrange matters so that the question of the existence of this effect could be answered with certainty?
>
> If the effect exists, then it is the end of our universal time, of the parallelogram of velocities, and of the whole of kinematics. I hope the effect does not exist—that would be much nicer. But since this is a genuine *experimentum crucis*, and since you may perhaps, despite the great difficulty, find a way to realize the matter, I wished to communicate it to you. (Ritz to Paschen, Jun. 14, 1908; Ritz, 1911b, 523–24)

The null result of the Michelson–Morley experiment could be explained equally well by assuming a ballistic character of light emission. The transverse Doppler effect, by contrast, cannot be accommodated within such an emission theory. Thus, Ritz was the first to characterize explicitly the transverse Doppler effect as an *experimentum* *crucis* for discriminating between his emission theory and Lorentz–Einstein relativistic kinematics itself, independently of more indirect consequences of the latter, such as the velocity dependence of the electron's mass. In all non-relativistic theories, including emission theories, a longitudinal Doppler effect can be accounted for, even though the underlying explanations differ. By contrast, only Lorentz–Einstein relativity predicts a genuinely relativistic Doppler component transverse to the direction of motion.[6]

After the meeting of the German Association of Natural Scientists and Physicians in Cologne in September 1908, many physicists felt compelled to favor relativity, driven both by Minkowski's (1909) ~~four dimensional~~ four-dimensional reformulation of the theory and by Bucherer's (1908) alleged experimental confirmation of the Lorentz–Einstein electron. However, Ritz's subsequent correspondence with his friend Ehrenfest suggests that he remained unconvinced (Staley, 2008, 268–271): "Minkowski forcefully promotes the principle of relativity *à la* Einstein. Hopefully this nonsense will disappear in 10 years. A wholly miserable makeshift [*Auskunfts-mittel*]. One could salvage any theory, no matter how false, in such a way" (Ritz to Ehrenfest, Dec. 17, 1908, cit. in ~~Staley, 2008, 270–271; translation modified~~ Staley, 2008, 270f.; translation modified). Rather than tinkering with the classical kinematical framework, Ritz (1908c) believed that many of the difficulties of contemporary physics, including the so-called ultraviolet catastrophe, could be resolved by reconstructing the foundations of electrodynamics through the abandonment of the ether and the exclusive use of retarded potentials.

Einstein (1909b) was the first to offer a systematic reply to this line of criticism, prompting Ritz's (1909) rejoinder; Ritz and Einstein (1909) ultimately agreed to disagree. Ritz maintained that irreversibility (the exclusion of the advanced potentials) must be imposed at the level of the fundamental electrodynamic laws, whereas Einstein held that it emerges only at the statistical level. In his March 1909 habilitation lecture on the principle of relativity in optics, Ritz (1911a) reiterated the stark dilemma facing physics at that time: (a) to preserve Maxwell electrodynamics and the wave theory of light with a source-independent velocity c at the price of adopting a new kinematics, or (b) to preserve the classical kinematical framework by radically modifying electrodynamics in the direction of an emission theory of light, in which fictitious light corpuscles would travel with velocities $c \pm v$. Two months after Ritz's death, in September 1909, Einstein, in his Salzburg lecture on the constitution of radiation (Einstein, 1909a), outlined, without mentioning Ritz, a contrasting program. Whereas Ritz sought to preserve classical kinematics and the continuity of radiation by abandoning the constancy of the speed of light, Einstein instead accepted the granular character of radiation and upheld the universal constancy of the speed of light by introducing a new kinematics.

4 Real and Apparent. Einstein's Last Remarks on the Transverse Doppler Effect

While struggling with the problem of radiation toward the end of 1909, Einstein (1910) wrote a review paper on relativity, published in French in two installments in

[6] Although the transverse Doppler factor can be retroactively read into Lorentz's equations, Lorentz never

the *Archives des sciences physiques et naturelles*. By his own admission, the paper contained nothing physically new (Einstein to Laub, Aug. 27, 1910; Einstein, 1987, Vol. 5, Doc. 224). Nevertheless, at least one conceptual novelty can be identified. Einstein explicitly addresses the question "What is a clock?" (Einstein, 1910, 21). Einstein famously defined a clock as any periodic system completely isolated from external influence that repeatedly passes through identical phases, such that—by the principle of sufficient reason—each period is taken to be identical to any other: "We therefore postulate that two identical phenomena have the same duration. The perfect clock thus defined plays, for the measurement of time, a role analogous to that of the perfect rigid body in the measurement of lengths" (Einstein, 1910, 21; fn. 1). After identifying the beginning and the end of a process, one can count the number N of identical cycles that occur. If the clock takes the form of a mechanism equipped with hands, completed cycles can be labeled by the integers $n = 1, 2, 3, \ldots$ via the positions of the hands. Assuming a stable intrinsic frequency of the clock ν_0 as a unit of time, this counting perspective can be translated into a measuring perspective. A clock at rest in a co-moving system directly measures the time coordinate of that system, such that the time coordinate of the nth tick is given by $\cancel{t_n = n/\nu_0}$$t'_n = n/\nu_0$.

One may then raise the separate question of whether systems exist in nature that realize the ideal of a perfect clock, that is, systems that ensure the physical stability of the frequency and therefore of the time unit. In classical physics, two objects could never be identical in every respect, since a complete physical description would, in principle, require infinitely many parameters, allowing systems to differ in arbitrarily small details. This is precisely why Einstein's appeal to atomic or ionic processes in the section dedicated to time dilation is epistemologically significant (Pierseaux, 2003). Spectroscopy provides strong empirical evidence that atoms and ions of the same species exhibit the same characteristic frequencies when compared at rest, regardless of how they were previously accelerated~~: the number of oscillations N is the same for the identical emission process~~. Thus, these systems ~~therefore~~ realize, in nature and to a high degree of accuracy, Einstein's definition of perfect clocks: "Since the oscillatory phenomena that give rise to a spectral line must be regarded as intra-atomic phenomena whose frequency [ν_0] is determined solely by the nature of the ions, we may take these ions to serve as clocks" (Einstein, 1910, 132–33).

The proper frequency ν_0 of the oscillatory motion of the ions provides a means of measuring time t: "this frequency will be known if one observes the spectrum produced by ions of the same nature, but at rest with respect to the observer" (Einstein, 1910, 133). Ions can be accelerated in canal rays by electric fields. Einstein clearly believed that there was sufficient empirical evidence to justify the assumption that acceleration does not permanently affect ions, which were expected to exhibit the same spectral lines when measured by a co-moving spectrograph. If this assumption is granted, the theory predicts that "there is an influence of the motion on the source, which reduces the *apparent* [*apparente*] frequency of the ion" (Einstein, 1910, 133; my emphasis), as measured by a spectrograph when the motion of the source is perpendicular to the line of sight.

identified, interpreted, or proposed it as a physical effect. The effect is not mentioned in Lorentz's (1916) 1909 Columbia lectures.

As suggested above, the use of the term *apparent* is likely intended to avoid confusion with Stark's interpretation of this 'influence' as a *real* dynamical contraction of the atomic frequency ν_0, that is, a reduction in the number of periods completed during the same time interval in the same frame. In Einstein's reading, by contrast, clocks retain the same intrinsic frequency ν_0. Since there is no relative motion along the line of sight, the observed frequency shift $\nu = \gamma^{-1}\nu_0$ can only be due to the fact that the same number N of periods is associated with different time intervals in different inertial frames:

$$\nu = \frac{N}{\Delta t} = \frac{N}{\gamma \Delta t'} = \frac{\nu_0}{\gamma} = \nu_0\sqrt{1 - \frac{v^2}{c^2}}.$$

~~The Lorentz factor thus appears~~ Thus, the Lorentz factor enters the expression not because of a 'real' reduction ~~in the number N of oscillations occurring during the same time interval. On the contrary, the premise of the argument is that N is the same in all inertial frames.~~ of the emitter's intrinsic frequency ν_0. The frequency shift $\nu = \gamma^{-1}\nu_0$ arises because, in a ~~non co-moving~~ non-co-moving frame K, ~~this same N is~~ the same completed oscillations are referred to a longer coordinate time interval, $\Delta t = \gamma \Delta t'$. The oscillations are therefore less densely packed in time in the ~~non co-moving~~ non-co-moving frame, which manifests itself as a redshift without any change in the intrinsic emission process given by $\nu_0 = N/\Delta t'$. In this sense, the effect is 'apparent'~~, since it disappears entirely in the co-moving frame K'~~: the shifted line is not detected by a spectrograph at relative rest with respect to the moving atom, which records the proper frequency instead.

In the ensuing months, following the debate raised by Ehrenfest's paradox, Einstein realized that the terms 'apparent' and 'real' gave rise to misunderstandings (Einstein to Varićak, Feb. 24, 1911; Einstein, 1987, Vol. 5[10], Doc. 255a). Some regarded kinematical effects, such as 'time dilation', as mere coordinate effects and therefore as 'apparent'. By contrast, Stark's 'frequency contraction' was taken to be 'real' ~~in~~ as a dynamical effect. To avoid confusion, Einstein suggested disentangling the conceptual pair real/apparent from kinematical/dynamical (Einstein, 1911). Relativistic time dilation, although a kinematical effect, is both apparent and real. It is 'apparent', since ~~it disappears for a suitably chosen~~ a co-moving ~~observer~~clock or spectrograph records the corresponding proper quantity; yet it is *real*, since it ~~cannot be eliminated for all non co-moving observers at once.~~ can be tested empirically. Given a suitable transverse observational arrangement, any spectrograph in relative motion with respect to the emitting ion will measure a shifted line; a spectrograph co-moving with the ion will measure the proper line. The 'apparent' change of frequency is nevertheless 'real' in the sense that it would be absent in a rival theory.[7] As Ehrenfest pointed out, this is precisely the reason why, in his letter to Paschen, his friend Ritz[8] proposed the transverse Doppler effect

[7]This case is by no means an *unicum* in the history of physics. A useful comparison can be drawn with stellar aberration in classical ether theory, which is likewise both *apparent* and *real*. It is apparent insofar as it disappears for a suitably chosen observer, yet real insofar as it ~~gives rise to systematic,~~ is empirically detectable~~effects that cannot be eliminated for all observers at once~~. Over the course of the year, astronomers must therefore inevitably reorient their telescopes toward the *apparent* position of the stars, whose annual displacement traces ellipses, although, of course, their *real* positions have remained unchanged. Nevertheless, stellar aberration is undeniably a genuine physical effect that would not be expected in a theory of complete ether drag..

[8]The letter was later published in the 1911b edition of Ritz's collected works.

as an *experimentum crucis* to decide between classical and relativistic kinematics[9]: "Whereas Einstein's theory *unconditionally* demands the existence of this effect [...][,] Ritz's emission theory *unconditionally* demands its nonexistence" (Ehrenfest, 1912, 319; my emphasis).

Conclusion

As is well known, Einstein repeatedly insisted that the new kinematics is not merely a matter of the conventional synchronization of clocks. Once established, the Lorentz transformations are either true or false in the sense that they can be *confirmed* or *disconfirmed* ~~through~~ by their empirically accessible consequences, provided that the coordinates are interpreted as quantities measurable by means of 'good' rods and clocks. ~~The~~ If atoms, the emitters of spectral lines, are such good clocks that they reliably register the time coordinate in their co-moving frame, then the transverse Doppler effect is indeed one such consequence~~, and at the time~~. At the time, it was the only one ~~feasibly to~~ that could feasibly be detected. Einstein's line of argument, disentangled from the historical details, may be reconstructed as follows:

1. *Empirical premise (spectral identity of atomic clocks at rest).* Atoms (or ions) of the same species, when compared at relative rest, exhibit the same intrinsic frequency ν_0, as evidenced by the reproducibility of their spectral lines.
2. *Physical premise (spectral identity of atomic clocks in uniform motion).* Uniform translational motion does not alter the ~~internal~~ intrinsic oscillatory process responsible for emission. Hence ~~, the number N of oscillations associated with a given emission process is the same in all non-accelerated inertial frames.~~identical ions, when compared at rest, possess the same intrinsic frequency $\nu_0 = N/\Delta t'$, independently of their state of uniform motion[10]
3. *Experimental restriction (transverse configuration).* The observation is arranged so as to eliminate longitudinal Doppler components, that is, any contribution due to motion along the line of sight.
4. *Prediction.* A Doppler shift in the frequency $\nu_0 = N/\Delta t'$ (for an atom at rest in ~~K'~~ K') persists even in the transverse case, namely

$$\nu = \gamma^{-1}\nu_0.$$

5. *Explanation.* ~~Since the number N of oscillations associated with a given emission process is invariant, the is entirely a consequence of the relativistic transformation of time, that is , of time dilation~~The same emission process involves the same count N of completed oscillations. What changes is not this count, but the time interval with respect to which it is evaluated: the same N is referred to different

[9] In the absence of a direct confirmation of the light postulate, which would soon be provided by de Sitter (1913).

[10] This premise corresponds to what is often called *boostability* (Brown, 2005, 30, 81, 121). As discussed above, Einstein made this assumption explicit as a criterion for 'good' rods and clocks in ~~Einstein, 1908, 420; fn. 1~~ Einstein, 1908, 420; fn. 1 and ~~126; fn. 2 Einstein, 1910~~126; fn. 2 Einstein, 1910..

time intervals in the two inertial descriptions:

$$\Delta t = \gamma\,\Delta t'.$$

In this form, the argument amounts at most to a provisional compromise. Whereas premise 3 is a definitional restriction built into the experimental arrangement, Einstein did not yet possess a genuine theoretical account of why atoms or ions ~~behave~~ possess stable, motion-independent proper frequencies in accordance with premises 1 and 2. At this stage, he could rely only on the overwhelming empirical evidence provided by spectroscopy.

It is well known that, beginning in the 1920s, Einstein increasingly acknowledged that a dynamical explanation of the periodic processes used as clocks would ultimately be required ~~(Einstein, 1921, 1923)Einstein1924,Einstein1926,Einstein1949~~ (Einstein, 1921, 1923, 1924, 1926). From a more fundamental standpoint, ~~clocks may~~ atomic clocks should be regarded as particular solutions of the underlying dynamical equations, with their behavior fixed by invariant parameters of the theory. Such parameters determine characteristic frequencies,[11] which in turn set the intrinsic periodicity of the admissible solutions. If the fundamental theory is Lorentz invariant, the same periodic solutions must occur in all ~~co-moving~~ inertial systems, with the ~~*same*~~ same proper frequency ν_0, whether atomic or otherwise. ~~In the case of atoms,~~

If such a theory ~~would~~ were available, premises 1 and 2 would no longer be merely assumed but would be derived as consequences of the theory itself. Such a theory would thus provide a dynamical explanation of the spectral identity of atoms—the empirical fact that identical atoms, when compared at relative rest, possess the same intrinsic frequency $\nu_0 = N/\Delta t'$ ~~and emit the same spectral lines when measured by a spectrograph co-moving with them in the~~ as measured in the co-moving frame K'.

~~If such a theory were available, premises 1 and 2 would no longer be merely assumed but would be derived as consequences of the theory itself.~~ Since premise 3 is merely definitional, there would then be no way to avoid conclusion 5, namely that prediction 4, the transverse Doppler effect, is ~~genuinely~~ an *apparent* coordinate effect:

$$\nu = \frac{N}{\gamma \Delta t'} = \frac{\nu_0}{\gamma} = \nu_0 \sqrt{1 - \frac{v^2}{c^2}}. \tag{4}$$

The transverse Doppler effectappears because the same number N of completed oscillations of the same process is spread over a longer time $\Delta t = \gamma \Delta t'$ in the non-co-moving system K. Thus, somewhat paradoxically, once a genuine dynamical explanation ~~of~~ for the behavior of clocks is ~~in place, one is forced to admit~~ established, one must concede that clock retardation admits only a purely ~~kinematical explanation: the appears because $\Delta t = \gamma\Delta t'$ in the non-co-moving system K. To avoid this conclusion one could deny~~ *kinematical explanation*. Frequency is nothing more than a ratio. If the numerator is held fixed, the observed frequency shift must be attributed to the denominator.

The only logically possible alternative is to hold the denominator fixed and attribute the same shift to the numerator instead. This would amount to denying

[11] For example, a mass parameter fixes the Compton frequency $\nu_C = mc^2/h$..

premises 1 and 2 and~~search for~~ , as Stark envisaged, searching for a dynamical explanation of the *real* ~~frequency~~ contraction of the ~~source.~~ intrinsic frequency ν_0:

$$\nu = \frac{\gamma^{-1}N}{\Delta T} = \frac{\nu_0}{\gamma} = \nu_0\sqrt{1-\frac{v^2}{c^2}}. \tag{5}$$

When the atom moves through the ether, the electromagnetic forces binding it are modified, so that the atom genuinely oscillates more slowly in the co-moving system K': the reduced number $\gamma^{-1}N$ of completed oscillations is spread over the same universal time ΔT. One thereby obtains a *dynamical explanation* of the observed frequency shift, but at the price of postulating an undetectable privileged frame K in which atoms oscillate at their natural frequency ν_0. Somewhat ironically, this ~~was precisely the strategy defended~~ approach was pursued by Herbert E. Ives (Ives and Stilwell, 1938) in connection with the experimental confirmation of the transverse Doppler effect in the late 1930s (Lalli, 2013). ~~However, Ives ' neo-Lorentzian move does not amount to a dynamical account that would *explain* relativistic kinematics in the sense advocated by modern dynamicists (Brown, 2005, vii); rather, it amounts to an ether-based dynamical conspiracy designed to *conceal* an underlying classical kinematics. At this point, however, it becomes unclear what a genuinely~~ Ives makes no mention of special relativity; instead, he presents his result as evidence that the change in frequency "predicted by the Larmor-Lorentz theory is verified" (Ives and Stilwell, 1938, 225; see Ives and Stilwell, 1941, 374). Unsurprisingly, Einstein responded rather tersely to Ives' announcement[12].

In the ensuing years, Einstein continued to concede that it was inconsistent to treat clocks "as independent physical objects" (Einstein to Swann, Jan. 24, 1942; Einstein Archives, n.d., 20-624) as he did in 1905. However, in confessing his youthful 'sin', Einstein (1949a, 61) was clearly not calling for a dynamical explanation of time dilation ~~would amount to.~~

~~In my view, the historically motivated case for the dynamical approach ultimately rests on a misunderstanding. Einstein's plea for a dynamical theory of rods and clocks has often been interpreted as a demand for *explanation* of the new kinematics, whereas it was primarily motivated by~~ *à la*Ives (1947). Rather, his concern was the problem of its *confirmation*. The ~~Lorentz transformations define the kinematical structure of special relativity. In order to test them empirically, one must employ rods and clocks as linked to the coordinate system. Yet rods and clocks are themselves physical systems governed by dynamical laws, which are in turn required to invariant with respect to the~~ transverse Doppler effectconfirms the Lorentz transformation of the time coordinate only *if* atoms are assumed to be 'good' clocks that, once placed at rest in an inertial frame under identical physical conditions, always tick at the same rate, *i.e.*,emit the same spectral lines. However, special relativitycannot prove whether this is the case. Strictly speaking, clocks should emerge as "solutions of the basic equations" of some fundamental theory (Einstein, 1949a, 59). Yet if such a theory were available, it would be necessary to reconsider what it means to provide 'empirical confirmation' of a prediction of the Lorentz transformations. The latter would then be tested

[12] Ives to Einstein, Jun. 24, 1938; Einstein Archives, n.d., 53-549; Einstein to Ives, Aug. 4, 1938; Einstein

using clocks that are themselves solutions of a Lorentz-invariant dynamical theory. Relativistic kinematics could no longer be confronted with experience independently of Lorentz-invariant dynamics. Rather, only the empirical success of the latter could provide indirect evidence for the Lorentz transformations. ~~For this reason~~

From the standpoint of confirmation, kinematics and dynamics ~~, from a strictly logical point of view, cannot be confronted with~~ would not confront experience separately, but rather, as Einstein frequently noted, only as a whole ~~.~~[13] ~~From the point of view of explanation~~that "in its entirety validates itself" (Einstein, 1949b, 678). From the standpoint of *explanation*, however, the ~~division of labour within this~~ separation between kinematics and dynamics within that whole remains intact. ~~The paper argues that the case of the illustrates this point particularly clearly. Lorentz invariance of the~~ Lorentz-invariant laws of matter ~~provides~~ provide a *dynamical* explanation of why all atomic clocks of the same kind ~~emit the same spectral lines at relative rest;~~ possess the same proper frequency ν_0 in all inertial systems. As we have seen, once this premise is established, the observed transverse shift of spectral lines admits ~~only~~ *only* a purely *kinematical* explanation as a time-coordinate effect~~: the same emission process is described using different time coordinates in different inertial frames. This difference has testable consequences that would not arise if the old kinematics held~~. Any further dynamical explanation within special relativitywould lack an independent explanandum. The case of the transverse Doppler effectshows with particular clarity that, from a historical point of view, Einstein's call for a dynamical explanation of the behavior of clocks was not intended to be a call for a dynamical explanation of time dilation. From a theoretical point of view, however, the pursuit of such an explanation within special relativitycan, of course, proceed on a different basis, without appealing to Einstein's authority[13].

Archives, n.d., 80-269.

[13]~~In this case, relativistic kinematics is not tested independently by observing rods and clocks, but rather indirectly through its role in guiding the construction of empirically adequate dynamical theories, such as relativistic quantum field theory. In the latter, the rest energy mc^2 fixes a fundamental energy scale and is associated, via Planck's relation, with the Compton frequency $\nu_C = mc^2/h$, thereby providing a dynamical basis for the stability of atomic spectral lines that can serve as clocks. The is therefore still expected as a purely coordinate effect, since the invariance of ν_0 is . In this sense, it serves primarily as a test of the Lorentz invariance of quantum field theory, rather than of relativistic kinematics in isolation. It may be that what dynamicists have in mind is precisely that the is *explained* by the Lorentz invariance of relativistic quantum field theory. However, the specific details of the theory are largely irrelevant: any Lorentz-invariant theory of matter that secures a frame-independent intrinsic frequency scale would suffice. It is the *requirement* of Lorentz invariance that does the explanatory work (Lange, 2016)~~.

[13]As is well known, John S. Bell (1976) suggested a third alternative between Einstein and Larmor dilation. One can in principle recover a relativistic effect such as the transverse Doppler effectby calculating how the physical forces within a moving atom are modified on the basis of a Lorentz-covariant theory formulated relative to a single chosen *inertial* frame, namely, the one in which the spectrograph is at rest. Bell's *pedagogical* intent was to provide an effective *illustration* that relativistic effects can be reconstructed dynamically within any chosen inertial frame. In the *philosophical* literature, it has been argued that Bell's proposal amounts to a genuine dynamical *explanation* of such effects (Brown and Pooley, 2001 ; Brown, 2005, sec. 1.4). However, it is important to note what this approach would entail. Since every inertial frame treats its co-moving atoms as oscillating at the proper frequency ν_0, Bell's proposal generates not one but infinitely many different equally legitimate, frame-relative dynamical explanations of one and the same observed frequency shift ν. The 'pre-established harmony' among these explanations ultimately rests on Bell's assumption that *all* fundamental interactions are Lorentz invariant (Bell, 1976, 143). The crucial question in the dynamical/kinematical debate is ultimately how this '*all*' is to be understood: as a remarkable coincidence among otherwise independent available dynamical laws, or as a requirement imposed

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